

Renal Acidification Mechanisms

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CHAPTER OUTLINE

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KEY POINTS

- Proximal tubule filtered bicarbonate reabsorption involves apical H^+ secretion by NHE3 and H^+ -ATPase, and in the neonatal kidney NHE8 substitutes for NHE3.
- Proximal tubule bicarbonate reabsorption is regulated by peritubular HCO_3^- and CO_2 , but not directly by peritubular pH.
- Proximal tubule basolateral NBCe1 is necessary for bicarbonate reabsorption and regulates both ammonia metabolism and citrate reabsorption.
- Aquaporins transport CO_2 and NH_3 in addition to H_2O .
- Renal ammonia transport involves selective transport of NH_3 and NH_4^+ by specific membrane proteins that exhibit significant axial and apical versus basolateral plasma membrane heterogeneity along the nephron and collecting duct.
- Ammonia metabolism involves both ammonia generation (ammoniogenesis) and ammonia recycling; the latter occurs through the protein glutamine synthetase.
- Renal interstitial sulfatides, probably by reversibly binding interstitial NH_4^+ , are necessary for normal ammonia metabolism.
- The bicarbonate secreting anion exchanger, pendrin, which is necessary for recovery from metabolic alkalosis, has a critical role in volume homeostasis and blood pressure regulation through roles both as a Cl^- reabsorbing protein and through indirect interactions with the Na^+ -reabsorbing protein, ENaC.

Maintaining normal acid–base homeostasis is critical for normal health. Acid–base disorders can lead to a number of clinical problems, such as growth retardation, nausea, and vomiting; increased susceptibility to cardiac arrhythmias; decreased cardiovascular catecholamine sensitivity; bone disorders, including osteoporosis and osteomalacia; recurrent nephrolithiasis, skeletal muscle atrophy; paresthesia; and coma.²⁶⁶ In people with chronic kidney disease (CKD), metabolic acidosis leads to more rapid progression of worsened renal function and increased risk of requiring renal replacement therapy.^{97,216} Finally, the presence of either metabolic acidosis or metabolic alkalosis correlates with increased mortality in patients both with and without CKD.^{214,289,327}

Acid–base homeostasis involves two separate but related processes, bicarbonate reabsorption and new bicarbonate generation. The first relates to the reabsorption of bicarbonate

filtered by the glomerulus. The second relates to the need to generate “new bicarbonate” to replenish bicarbonate that neutralizes endogenous and exogenous fixed acid loads. Finally, a number of pathophysiologic conditions generate acid or alkali loads to which the kidneys must respond to in order to maintain acid–base homeostasis.

BICARBONATE REABSORPTION

Bicarbonate reabsorption involves coordinated transport events in multiple nephron segments (Fig. 9.1). The proximal tubule reabsorbs the majority of filtered bicarbonate. Little-to-no bicarbonate reabsorption occurs in the thin descending limb of the loop of Henle, moderate reabsorption occurs in the thick ascending limb (TAL) of Henle loop, and the

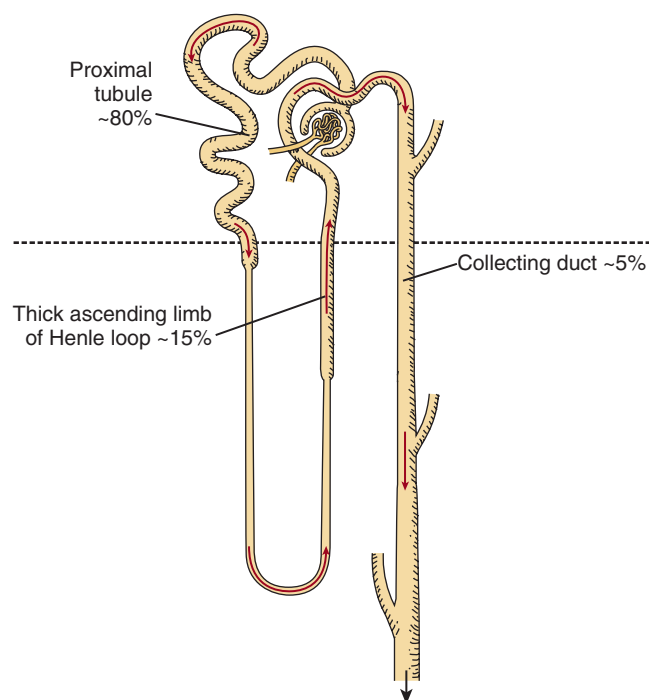


Fig. 9.1 Summary of sites of bicarbonate reabsorption. The proximal tubule is the primary site quantitatively for filtered bicarbonate reabsorption. Minimal reabsorption occurs in the thin limb of the loop of Henle. The thick ascending limb of the loop of Henle reabsorbs the majority of the bicarbonate not reabsorbed in the proximal tubule. The collecting duct is the primary site for reabsorption of the remaining filtered bicarbonate.

remaining filtered bicarbonate is reabsorbed in the distal convoluted tubule (DCT), connecting segment (CNT), initial collecting tubule (ICT), and the collecting duct.

PROXIMAL TUBULE

GENERAL TRANSPORT MECHANISMS

Proximal tubule bicarbonate reabsorption involves several distinct, but interconnected, processes (Fig. 9.2). First, protons (H^+) are secreted into the luminal fluid. Multiple proteins mediate H^+ secretion; the apical Na^+/H^+ exchanger, NHE3, and an apical H^+ -ATPase are the primary mechanisms of proton secretion in the adult kidney. In the neonatal kidney, the Na^+/H^+ exchanger, NHE8, appears to substitute for NHE3 as the primary Na^+/H^+ exchanger (NHE) isoform.³⁰ In the adult kidney, NHE3 is responsible for 60%–70% of H^+ secretion and H^+ -ATPase accounts for the majority of the remainder.

Secreted H^+ combines with luminal HCO_3^- to form carbonic acid (H_2CO_3). Luminal carbonic acid dissociates to water (H_2O) and carbon dioxide (CO_2). Although this can occur spontaneously, the spontaneous dehydration rate is inadequate to support normal rates of proximal tubule bicarbonate reabsorption. The dehydration reaction is catalyzed by carbonic anhydrase IV (CA IV), a membrane-bound carbonic anhydrase isoform present in the proximal tubule brush border.

Luminal CO_2 then moves across the apical plasma membrane into the cell. Although this process has traditionally

been thought to occur through lipid-phase diffusion, the integral membrane protein, aquaporin 1 (AQP1), may mediate ~50% of CO_2 transport across the apical plasma membrane.⁴⁶ Cytosolic CO_2 is then hydrated, forming carbonic acid, through a process accelerated by the cytosolic carbonic anhydrase, carbonic anhydrase II (CA II). Cytosolic carbonic acid spontaneously dissociates to H^+ and HCO_3^- . This “replenishes” the H^+ secreted across the apical plasma membrane by apical NHE3 and H^+ -ATPase.

Cytosolic HCO_3^- is transported across the basolateral plasma membrane. In the S1 and S2 segments of the proximal tubule the primary HCO_3^- transport mechanism is a sodium-coupled, electrogenic bicarbonate cotransporter, NBCe1-A.^{1,256} Because NBCe1-A is electrogenic, generation and regulation of the transmembrane voltage between cytoplasm and interstitium is important, and appears to be related to extracellular pH-dependent activation of the basolateral TWIK-related acid-sensitive K^+ channel, TASK2.⁴⁴⁵ In the S3 segment, an Na^+ -dependent, Cl^-/HCO_3^- exchanger appears to be the primary mechanism of basolateral HCO_3^- transport,²¹⁷ although NBCe1 may also contribute.²⁸⁴

In addition to active H^+ secretion-mediated luminal bicarbonate reabsorption, the proximal tubule also exhibits passive H^+ and bicarbonate transport. Because bicarbonate reabsorption decreases the luminal bicarbonate concentration and increases the luminal H^+ concentration relative to the peritubular space, passive bicarbonate transport results in bicarbonate secretion, which limits net bicarbonate reabsorption. The molecular mechanisms of bicarbonate backleak are unclear, but several functional aspects are known. It is quantitatively less in the newborn than in the adult kidney,³²³ is decreased by angiotensin II (AngII),²⁴² involves both paracellular and transcellular components, and involves membrane proteins, but not NHE3.^{158,315} Quantitatively, the bicarbonate backleak rate is less than that of bicarbonate reabsorption in the initial portions of the proximal tubule. However, in more distal portions, particularly when luminal bicarbonate concentrations have decreased as a result of more proximal bicarbonate reabsorption, bicarbonate backleak rates are greater as a result of the greater transepithelial bicarbonate gradient. Simultaneously, the lower luminal pH limits NHE3-mediated proton secretion and bicarbonate reabsorption. This can result in the bicarbonate backleak rate becoming equivalent to the bicarbonate reabsorption rate. When this occurs, there is no further net bicarbonate reabsorption. Under typical circumstances, this occurs when luminal bicarbonate concentrations have decreased to approximately 6 mmol/L, corresponding to a luminal pH of 6.8.

Proteins Involved in Proximal Tubule Bicarbonate Reabsorption

Na^+/H^+ Exchangers. Na^+/H^+ exchangers are expressed widely in the kidney, where they function in intracellular pH regulation, transepithelial bicarbonate reabsorption, and vacuolar acidification. All use the extracellular-to-intracellular H^+ gradient to enable secondary active, electroneutral H^+ secretion. Although the preferred ions are Na^+ and H^+ , Li^+ can substitute for Na^+ and NH_4^+ can substitute for H^+ .²⁰⁴ The latter process, which enables Na^+/NH_4^+ exchange, appears to be important for proximal tubule NH_4^+ secretion.²⁷⁴

NHE3 (SLC9A3) is the primary apical Na^+/H^+ exchanger in the proximal tubule and mediates the majority of

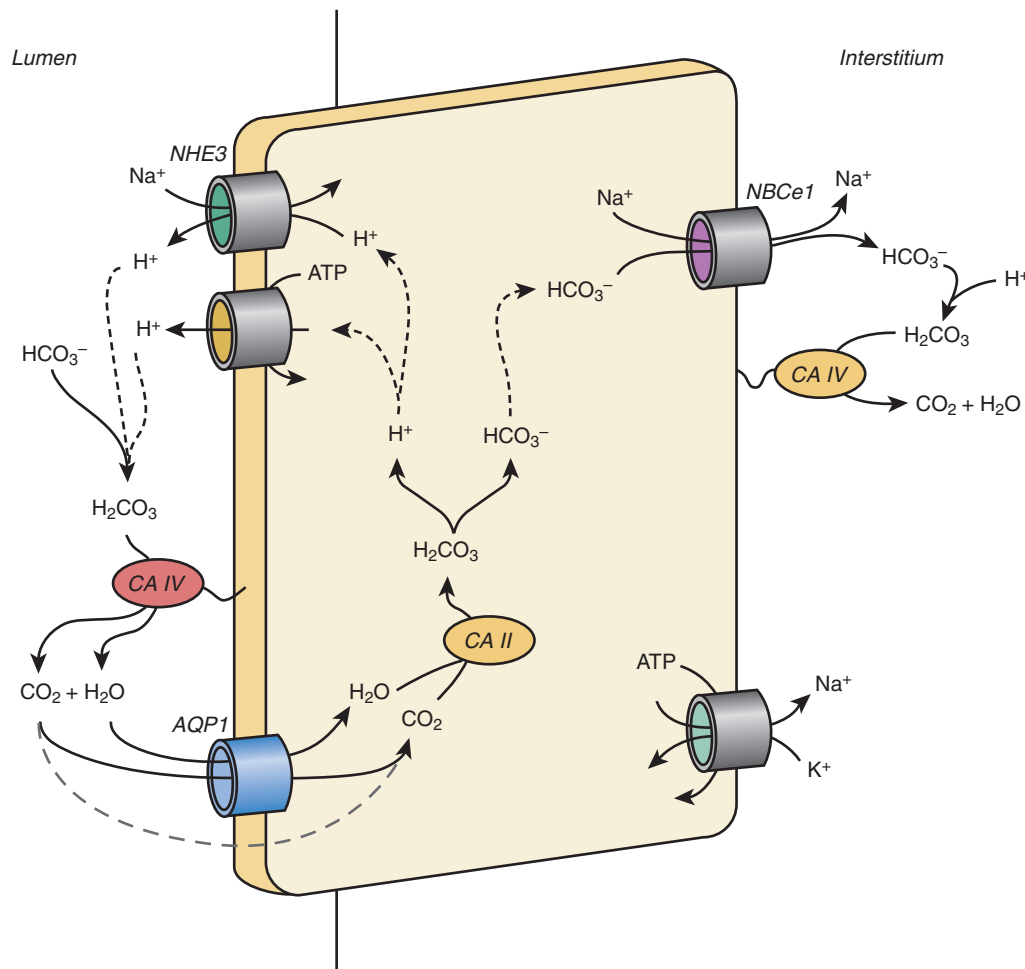


Fig. 9.2 Bicarbonate reabsorption in the proximal tubule. Proximal tubule HCO_3^- reabsorption involves integrated function of multiple proteins. Protons are secreted by the Na^+/H^+ exchanger, *NHE3*, and by H^+ -ATPase, and titrate luminal HCO_3^- to H_2CO_3 . Luminal H_2CO_3 dehydration to H_2O and CO_2 is accelerated by luminal carbonic anhydrase activity mediated by *CA IV*. CO_2 enters the cell via aquaporin I (*AQP1*) and most likely also via passive lipid-phase diffusion, where its hydration to H_2CO_3 is accelerated by cytoplasmic *CA II*. H_2CO_3 rapidly dissociates to H^+ and HCO_3^- , thereby “replenishing” the secreted cytosolic H^+ . Cytosolic HCO_3^- exits across the basolateral plasma membrane primarily by the electrogenic sodium-bicarbonate cotransporter, *NBCe1-A*. In the PST, a basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchange activity (not shown) is the primary basolateral HCO_3^- exit mechanism. *ATP*, Adenosine triphosphate; *CA IV*, carbonic anhydrase 4; *PST*, proximal straight tubule.

luminal bicarbonate reabsorption. Multiple mechanisms regulate *NHE3*; the best studied are parathyroid hormone (PTH), dopamine, and AngII. Both PTH and dopamine inhibit *NHE3* activity, whereas AngII has a biphasic effect, stimulatory at low concentrations and inhibitory at high concentrations. Both PTH and dopamine increase intracellular cyclic adenosine monophosphate (cAMP) levels, leading to decreased *NHE3* activity,⁸⁷ and dopamine also has protein kinase C- (PKC)-dependent effects.¹³⁷ AngII decreases cAMP levels and activates PKC, tyrosine kinase, and phosphatidylinositol-3-kinase.¹⁷⁸

NHE3 phosphorylation is an important regulatory mechanism. Serine-552 (in the rat sequence) is a consensus protein kinase A (PKA) phosphorylation site, and phosphorylation of this site causes localization to the coated pit region of the brush-border membrane, where *NHE3* cannot contribute to bicarbonate reabsorption.²¹¹ Similarly, phosphorylation of serine-719 regulates insertion into the plasma membrane.³⁴⁵ This phosphorylation affects interactions of multiple signaling proteins with *NHE3*, alters the size of the *NHE3* protein

binding complex, and alters *NHE3* lipid raft distribution, which cause changes in specific aspects of basal as well as acutely regulated *NHE* exchange activity.³⁴⁶ Dephosphorylation, mediated by the serine/threonine phosphatase PP1, but not PP2, at serines 552 and 605, and at other novel phosphorylation sites, stimulates *NHE3* activity.¹⁰⁵ Ang II counteracts the effects of cAMP/PKA by dephosphorylating *NHE3* at serine 552, which may be a key event in the regulation of renal proximal tubule sodium handling.⁸⁹

Movement of *NHE3* between different subcellular locations, including microvilli, intermicrovillar clefts, endosomes, and the cytoplasm, is an important regulatory mechanism. Only *NHE3* in microvilli contributes to bicarbonate reabsorption. Redistribution within these domains is regulated by a variety of factors, including renal sympathetic nerve activity, glucocorticoids, insulin, AngII, dopamine, and PTH.^{22,39,87,260,334} This process involves a number of cellular proteins, including dynamin, NHERF-1, clathrin-coated vesicles, calcineurin homologous protein-1, ezrin phosphorylation, G-protein alpha subunits, and G-protein beta-gamma dimers.^{8,100,179}

NHE8 (SLC9A8) is a second Na^+/H^+ exchanger found in the proximal tubule.¹⁴⁹ Under normal conditions in the adult kidney, NHE8 is mostly intracellular,³⁸ but in the absence of NHE3 and during acid-loading in normal mice, NHE8 protein expression increases in brush border membrane fractions and contributes to bicarbonate reabsorption.³⁰ In the neonatal kidney, brush border membrane NHE8 expression is increased and NHE3 expression is decreased compared with adult kidneys, suggesting that NHE8 is the primary mechanism of apical NHE activity in the neonatal kidney.^{30,31,309,406}

NHE1 (SLC9A1) is a third member of the NHE family that is present in the proximal tubule. NHE1 is a ubiquitous sodium hydrogen exchanger present in essentially all cells in mammalian organs and is located in the proximal tubule in the basolateral plasma membrane.³⁴ Its role in the proximal tubule appears to be acute intracellular pH regulation.⁶⁰

H^+ -ATPase. A second mechanism of proximal tubule apical H^+ secretion involves the vacuolar H^+ -ATPase.⁴⁴³ H^+ -ATPase is expressed in the brush border microvilli, the base of the brush border, and apical invaginations between clathrin-coated domains.⁵² H^+ -ATPase also acidifies proximal tubule endosomes and lysosomes, senses endosomal pH, and is involved in recruiting trafficking proteins to acidified vesicles, thereby ensuring appropriate progression from early endosomes to lysosomes.⁵⁴ Proximal tubule H^+ -ATPase activity is increased by AngII, increased axial flow, and chronic metabolic acidosis.^{74,103,427} H^+ -ATPase has a direct binding interaction with aldolase, which may underlie the development of proximal RTA in individuals with hereditary fructose intolerance.²⁴⁷ In addition, PKA stimulates and the adenosine monophosphate-activated protein kinase (AMPK) inhibits apical plasma membrane H^+ -ATPase insertion and activity.⁷

NBCe1 (SLC4A4). Basolateral bicarbonate exit largely is mediated by the electroneutral sodium-bicarbonate cotransporter, NBCe1. In humans, three splice variants of the NBCe1 gene are known; NBCe1-A, also known as kNBC1, is the primary splice variant expressed in the kidney, where it is found exclusively in the basolateral plasma membrane in the proximal convoluted tubule.^{61,256} In mice, there are a total of five known splice variants.²¹⁹ NBCe1-A has large cytoplasmic amino- and carboxy-termini tails, 14 transmembrane domains, and two glycosylation sites.^{45,244,335,490}

NBCe1-A in the proximal tubule mediates the coupled net movement of Na^+ and HCO_3^- . The majority of evidence suggests this involves a 1:3 ratio of Na^+ and HCO_3^- equivalents.^{151,336} Because the cytoplasm is negatively charged relative to the peritubular compartment, this electrical gradient provides the driving gradient to enable the coupled movement of Na^+ and HCO_3^- out of the cell, against their concentration gradient. The coupling ratio of Na^+ and HCO_3^- is likely to be critically important: a 1:3 coupling mediates net HCO_3^- efflux, whereas with a 1:2 ratio, depending on the assumptions of intracellular Na^+ and HCO_3^- concentration and of basolateral membrane voltage, the net electrochemical gradient may favor HCO_3^- influx. Indeed, some proximal renal tubular acidosis (RTA) cases may result from NBCe1 mutations that alter the coupling ratio.⁴⁹¹ However, the specific molecular mechanisms of this 1:3 coupling ratio are only partially understood. In nonrenal cells, NBCe1-A appears to have a

1:2 coupling ratio.^{151,173,232} Moreover, it appears that when NBCe1-A is expressed in a proximal tubule cell line, Ser-982 phosphorylation shifts the stoichiometry from 1:3 to 1:2.¹⁵² Two aspartate residues near Ser-982 are necessary for this stoichiometry shift.¹⁵³ Thus, the coupling ratio of 1:3 appears to be important for NBCe1-A to facilitate bicarbonate exit and to be determined, at least in part, by phosphorylation-dephosphorylation of specific amino acid residues.

Proximal tubule NBCe1-A-mediated bicarbonate transport is regulated by physiologic conditions. Transport activity increases in response to metabolic acidosis and to a variety of stimuli that increase bicarbonate reabsorption.^{130,314,373} However, changes in steady-state protein expression do not appear to be an important regulatory mechanism, as metabolic acidosis does not appear to alter NBCe1-A expression.²²⁰ Other factors known to regulate NBCe1-A activity include intracellular ATP, possibly through an as yet unidentified kinase,¹⁷³ and a regulated recycling pathway involving PKC³⁰⁵ and calcium/calmodulin-dependent protein kinase II.³⁰⁶ Although Ste20/SPS1-related proline-alanine-rich kinase (SPAK)-dependent Ser65 phosphorylation and IRBIT- and SPAK-dependent Thr49 phosphorylation regulate NBCe1-B activity,¹⁷⁶ as a result of alternative splicing these residues are not present in NBCe1-A. Moreover, because of differential splicing, the auto-inhibitory domain present in NBCe1-B and NBCe1-C is not present in NBCe1-A, which results in approximately fourfold greater basal activity of NBCe1-A compared with the -B and -C variants.²⁵⁹

Defects in NBCe1 are the most common cause of autosomal recessive proximal RTA (pRTA).^{186,219,491} In addition to causing severe pRTA, NBCe1 defects can cause growth and mental retardation, basal ganglia calcification, cataracts, corneal opacities (band keratopathy), glaucoma, elevated serum amylase and lipase, and defects in the enamel suggestive of amelogenesis imperfecta.^{186,219} In mice, homozygous NBCe1 deletion causes a very severe phenotype, with severe metabolic acidosis, marked volume depletion, and death within a few weeks after birth. Heterozygous deletion causes a milder phenotype but still causes development of pRTA.^{129,167}

NBCe1, in addition to its role in proximal tubule bicarbonate reabsorption, also has a critical role regulating other proximal tubule acid-base functions. NBCe1 gene deletion causes abnormal renal ammonia and organic anion metabolism, and these effects appear to be mediated through alterations in proximal tubule proteins involved in the metabolism and/or transport of these acid-base components.^{167,293} This critical role of NBCe1 appears to be due, at least in part, to the A splice variant, NBCe1-A. Recent studies show that mice with NBCe1-A-specific deletion live to adulthood, have spontaneous metabolic acidosis without increased ammonia excretion, an abnormal physiologic response, and have greatly impaired ammonia metabolism and excretion response to acid loading.²²⁸ The abnormal ammonia excretion correlates with abnormal expression of critical proteins involved in ammoniogenesis, including PDG, PEPCK, and GS.²²⁸

Carbonic Anhydrase. Carbonic anhydrases are a family of zinc metalloenzymes that catalyze the reversible hydration of CO_2 to form carbonic acid (H_2CO_3), reaction A in the equation:



In the absence of carbonic anhydrase, the hydration/dehydration reaction (reaction A) is rate limiting, whereas reaction B occurs essentially instantaneously.

CA II. CA II is the predominant carbonic anhydrase in the kidney and in the proximal tubule. It is located in the cytoplasm of the proximal tubule, in addition to multiple other sites in the kidney, including thin descending limb, thick ascending limb of the loop of Henle (TAL), and intercalated cells. In the mouse kidney, CA II is also expressed in collecting duct principal cells.

CA IV. CA IV is found in the proximal tubule and in intercalated cells in the collecting duct.³⁵² CA IV is linked to the plasma membrane via a glycosylphosphatidylinositol (GPI) anchor and extends into the extracellular compartment; the active site is thus extracellular, not intracellular.⁴⁹² In the proximal tubule, CA IV is expressed in both apical and basolateral plasma membranes where, by facilitating HCO_3^- interconversion with CO_2 , it contributes to transepithelial bicarbonate reabsorption.⁵³

REGULATION OF PROXIMAL TUBULE BICARBONATE REABSORPTION

Systemic Acid-Base

Changes in extracellular acid-base status profoundly alter proximal tubule bicarbonate reabsorption. Both metabolic and respiratory acidosis increase bicarbonate reabsorption, and alkalosis decreases it. This occurs with both acute and chronic pH changes, although the effects are substantially greater with chronic changes. It is important to note that these effects are mediated through changes in interstitial (i.e., peritubular) HCO_3^- , and pCO_2 . Changes in luminal HCO_3^- have the opposite effect on proximal tubule bicarbonate transport, a manifestation of glomerular-tubular balance.

Recent studies have begun to elucidate the mechanisms through which extracellular bicarbonate and CO_2 regulate proximal tubule bicarbonate reabsorption. Changes in either peritubular CO_2 or HCO_3^- concentration, but not pH when the other two components are constant, alter bicarbonate reabsorption.⁴⁸⁹ These effects are specific to bicarbonate reabsorption, as fluid reabsorption rates do not change. The basolateral plasma membrane protein, protein-tyrosine phosphatase, receptor-type, gamma (PTPRG), is necessary for this molecular sensing.⁴⁸⁸ PTPRG may couple to the ErbB tyrosine kinases, ErbB1 and ErbB2, as inhibitors of these proteins block these responses, and phosphorylation of ErbB1 and ErbB2 is regulated by both bicarbonate and CO_2 concentration.³⁷⁰ An additional mechanism may involve the intrarenal angiotensin system, as peritubular CO_2 stimulates intracellular AngII production and luminal secretion, which acts through an apical AT_1 receptor to stimulate bicarbonate reabsorption.^{485,487}

Chronic metabolic acidosis increases proximal tubule bicarbonate reabsorption more than acute metabolic acidosis. This adaptive increase involves increased NHE3 expression and activity and increased H^+ -ATPase activity,^{16,74,314} but not detectable changes in NBCe1 or NBCe1-A expression.^{220,228} Glucocorticoid levels rise with chronic metabolic acidosis,⁴⁶⁴ and glucocorticoid receptor activation enhances acidosis-induced increases in NHE3 expression and apical trafficking.¹⁴

Luminal Flow Rate. Renal bicarbonate reabsorption changes in parallel with glomerular filtration rate and luminal flow.³¹⁰

Increased luminal flow enhances apical plasma membrane NHE3 activity.³¹² In addition, increased flow minimizes changes in the luminal bicarbonate concentration, thereby maintaining a higher mean luminal bicarbonate concentration, which facilitates bicarbonate reabsorption.¹³ Proximal tubule brush border microvilli may function as flow sensors, with drag force transmitted through the actin filament, altering cytoskeletal elements and regulating transport.¹⁰³

AngII. AngII is an important regulator of proximal tubule ion transport, including bicarbonate reabsorption. Low AngII concentrations increase but high concentrations inhibit bicarbonate reabsorption.^{78,442} Both luminal and peritubular low-dose AngII stimulate bicarbonate reabsorption, mediated predominantly through apical and basolateral AT_1 receptors. Acidosis increases AT_1 receptor expression, which may contribute to adaptive changes in bicarbonate reabsorption.²⁷⁹

Potassium. Chronic hypokalemia stimulates and hyperkalemia inhibits proximal tubule bicarbonate reabsorption.³²⁸ This is associated with parallel changes in apical Na^+/H^+ exchange and basolateral sodium-bicarbonate cotransport activity,³⁷² and involves increased apical and basolateral plasma membrane AT_1 receptor expression.¹²⁶ Acute changes in extracellular potassium concentration, however, do not alter proximal tubule bicarbonate transport.⁷⁷

Endothelin. Endothelin has important and direct effects on ion transport in a variety of renal epithelial cells, including the proximal tubule. Endothelin can be produced in the proximal tubule and exhibits an autocrine effect to stimulate NHE3.²⁴⁰ In particular, metabolic acidosis-induced increases in NHE3 expression may require endothelin B (ET-B) receptor activation.²²²

PTH. PTH acutely inhibits proximal tubule bicarbonate reabsorption through activation of adenyl cyclase and increased intracellular cAMP production.²⁶³ Systemic PTH administration leads acutely to metabolic acidosis, but chronically leads to metabolic alkalosis.¹⁸³ The acute effect is due primarily to increased urinary bicarbonate excretion, likely due to changes in proximal tubule bicarbonate reabsorption; the chronic effect is due to increased titratable acid excretion, which is likely due to increased excretion of dihydrogen and hydrogen phosphate.¹⁸³

Calcium Sensing Receptor. The calcium sensing receptor (CaSR) is present in the apical membrane in the proximal tubule. CaSR activation, either by increased luminal calcium or through calcimimetic agents, increases bicarbonate reabsorption through a mechanism likely involving the activation of apical NHE3.⁶⁶ CaSR activation may modulate the effects of PTH in proximal tubule bicarbonate reabsorption; hypercalcemia resulting from excess PTH has the opposite effect of PTH alone on bicarbonate transport.

LOOP OF HENLE

The TAL of the loop of Henle reabsorbs ~15% of the filtered bicarbonate load. The overall schema is fundamentally similar to that in the proximal tubule. Apical Na^+/H^+ exchange and vacuolar H^+ -ATPase secrete H^+ . Quantitatively, apical Na^+/H^+ exchange activity is the major H^+ secretory mechanism;

vacuolar H^+ -ATPase activity is present, but has at most a minor role in bicarbonate reabsorption.^{67,148} Two Na^+/H^+ exchanger isoforms are present in the TAL, NHE2, and NHE3, and NHE3 appears to be the predominant isoform.^{409,444} Secreted H^+ reacts with luminal HCO_3^- , forming H_2CO_3 , which dissociates to CO_2 and H_2O . Whether luminal CA IV is present is unclear, with conflicting reports in the literature.^{53,402} Luminal CO_2 moves down its concentration gradient across the apical plasma membrane into the cell cytoplasm. Cytoplasmic CA II catalyzes CO_2 hydration to form H_2CO_3 , which dissociates to H^+ and HCO_3^- , thereby regenerating the H^+ secreted across the apical plasma membrane. Several basolateral bicarbonate exit mechanisms are present. These include basolateral Cl^-/HCO_3^- exchange, possibly AE2,¹¹ and a coupled $K^+-HCO_3^-$ cotransport activity that may be mediated by KCC4.²⁷⁰ Although an electroneutral sodium–bicarbonate cotransporter (NBCn1) is present,⁴²⁴ electrochemical gradients for its transport favor bicarbonate uptake, not extrusion, suggesting that it is unlikely to contribute significantly to basolateral bicarbonate exit.

Several plasma membrane proteins either directly or indirectly alter bicarbonate reabsorption. Inhibiting the apical $Na^+-K^+-2Cl^-$ cotransporter, NKCC2, increases bicarbonate reabsorption.⁶⁷ This may occur because inhibiting NKCC2 decreases Na^+ entry, which decreases intracellular Na^+ , increasing the Na^+ uptake gradient for apical Na^+/H^+ exchange and thereby increasing bicarbonate reabsorption. Inhibiting basolateral Na^+/H^+ exchange activity decreases bicarbonate reabsorption through cytoskeletal alterations that decrease apical NHE3 expression.^{145,447}

Regulation of TAL Bicarbonate Reabsorption

A variety of stimuli regulate TAL bicarbonate reabsorption. Metabolic acidosis increases TAL bicarbonate reabsorption,^{69,138} but whether the effects are specific to metabolic acidosis or due to other mechanisms is not clear. One study reported that metabolic acidosis induced with NH_4Cl and that chloride loading with $NaCl$ had similar effects on bicarbonate transport, raising the possibility that chloride loads, not acid loads, regulate TAL bicarbonate transport.¹³⁸ Data in favor of acidosis regulating TAL bicarbonate transport are that NH_4Cl -induced metabolic acidosis, but not equivalent chloride loading with $NaCl$, increases TAL NHE3 expression.¹⁹⁸ Further supporting a role of the TAL in acid–base regulation is that experimental models of metabolic alkalosis decrease bicarbonate reabsorption.¹⁴¹

Several hormones regulate bicarbonate reabsorption. AngII stimulates TAL bicarbonate reabsorption, likely through activation of AT_1 receptors.^{68,272} Glucocorticoid receptors are present in the TAL and glucocorticoids are necessary for normal bicarbonate reabsorption.⁴⁰⁸ Mineralocorticoids, at high concentrations, stimulate bicarbonate reabsorption,¹⁴¹ but their absence does not alter basal transport.⁴⁰⁸ Arginine vasopressin (AVP) inhibits bicarbonate reabsorption through prostaglandin E_2 -mediated inhibition of apical Na^+/H^+ exchange activity.^{44,139} PTH inhibits bicarbonate reabsorption, but the effect is less than the effect of AVP.¹³⁹

Cytokines also regulate bicarbonate transport. Lipopolysaccharide (LPS) inhibits transport; this effect involves the cytokine receptor, TLR4, and separate pathways activated by luminal and peritubular LPS. Luminal LPS involves the mTOR pathway, whereas peritubular LPS functions through the

mitogen-activated protein kinase kinase (MAP) extracellular signal-related kinase (ERK) pathway.^{146,448} High-mobility group box 1 (HMGB1) is a nuclear protein released extracellularly in response to infection or injury, where it interacts with toll-like receptor 4 (TLR4) and other receptors to mediate inflammation. It inhibits TAL bicarbonate reabsorption through a receptor for advanced glycation end products (RAGE)-dependent mechanism that involves signaling through Rho and Rho-associated kinase (ROCK).^{147,446}

Another important regulatory factor is medullary osmolality. Increased tonicity inhibits and decreased tonicity stimulates bicarbonate reabsorption; this occurs through phosphatidylinositol 3-kinase-mediated changes in apical Na^+/H^+ exchange activity.^{140,144} In addition, AVP, which contributes to the development of the medullary osmotic gradient, discussed elsewhere in this textbook, inhibits bicarbonate reabsorption.¹³⁹

ACID–BASE TRANSPORTERS IN THE TAL

Many of the major H^+ and HCO_3^- transporters were discussed earlier in relation to the proximal tubule and are not repeated here.

NBCn1 (SLC4A7)

NBCn1 facilitates the electroneutral, coupled transport of Na^+ and HCO_3^- in a 1:1 ratio. In the kidney, NBCn1 is found in the basolateral plasma membrane in the TAL, outer medullary collecting duct (OMCD) intercalated cells, and terminal inner medullary collecting duct (IMCD).^{220,311} Because the concentrations of Na^+ and HCO_3^- are generally lower in the cytoplasm than in the interstitium, basolateral NBCn1 likely mediates peritubular HCO_3^- uptake. Moreover, both metabolic acidosis and hypokalemia increase TAL NBCn1 expression.^{189,220} Thus, NBCn1 is unlikely to mediate a critical role in bicarbonate reabsorption. Instead, it is likely to contribute to ammonia reabsorption, which will be discussed later.

DISTAL CONVOLUTED TUBULE

The DCT consists of two cell types, DCT cells and intercalated cells, and the mechanisms involved in bicarbonate reabsorption appear to differ between DCT and intercalated cells. DCT cells express apical NHE2⁷⁵ and NHE2 inhibitors decrease bicarbonate reabsorption.⁴⁴⁴ Basolateral HCO_3^- exit likely involves AE2.¹¹ A basolateral Cl^- channel that has limited HCO_3^- permeability may also contribute.⁴⁶³ Cytosolic CA II is present, but not apical CA IV.⁵³ In the late DCT intercalated cells are present.²⁵¹ Quantitatively, intercalated cells constitute only a very small proportion of all cells in the DCT, ~4% and 7% in the mouse and rat kidneys, respectively.²⁰¹ The majority of intercalated cells in the DCT are type A and non-A, non-B intercalated cells.²⁰¹

COLLECTING DUCT

The renal collecting duct is the final site of bicarbonate reabsorption and both reabsorbs and secretes luminal bicarbonate.²⁶¹ Specific proteins in specific epithelial cell types, which vary in type and frequency in different collecting duct segments, mediate these processes.

Collecting Duct Segments

Technically, the collecting duct begins with the ICT, immediately distal to the CNT, and extends through the IMCD. The

CNT arises from a different embryonic origin than the ICT and the remainder of the collecting duct. However, the CNT is included in the discussion of the role of the collecting duct in acid–base regulation because it has cell types and acid–base transport mechanisms similar to the collecting duct. Different portions of the collecting duct are identified by where they reside: ICT, cortical collecting duct (CCD), outer medullary collecting duct in the outer stripe (OMCDo), outer medullary collecting duct in the inner stripe (OMCDi), and the IMCD.

Cell Composition

Collecting duct segments contain several distinct epithelial cell types, and the cellular composition differs in the various collecting duct segments. Two distinct cell types, intercalated cells and principal cells, are present. Principal cells account for ~60%–65% of cells and intercalated cells account for the remainder in the ICT, CCD, and OMCD. In the IMCD, the proportion of intercalated cells is less, about 10% of cells in the initial portion of the rat IMCD, and it decreases progressively from the outer medullary–inner medullary junction distally, completing disappearing by the middle of the papilla. In the terminal IMCD, the epithelium is composed of IMCD cells, a cell distinct from both intercalated cells and principal cells. The CNT contains both intercalated cells and a cell type specific to the CNT, termed the CNT cell; in some species, principal cells are also present.

At least three distinct intercalated cell subtypes exist: the type A (or α) intercalated cell, the type B (or β) intercalated cell, and the non-A, non-B intercalated cell (Fig. 9.3). In the CNT, both type A and non-A, non-B intercalated cells are present, and type B intercalated cells are infrequent. In the CCD, both type A and type B intercalated cells are present, and the non-A, non-B cell is infrequent. In the OMCD and IMCD, only the type A intercalated cell is present under normal conditions.

Type A Intercalated Cell

The type A intercalated cell is involved in H^+ secretion, HCO_3^- reabsorption, and ammonia secretion. The proteins involved in these processes are, in general, different from those in the proximal tubule and TAL (Fig. 9.4).

Both vacuolar H^+ -ATPase and P-type H^+ - K^+ -ATPases are involved in apical H^+ secretion. H^+ -ATPase is abundant in the apical plasma membrane and in apical cytoplasmic tubulovesicles in type A intercalated cells. H^+ -ATPase undergoes trafficking between the cytoplasmic compartment and the apical plasma membrane; this mechanism, rather than changes in total protein expression, appears to be the major adaptive response to acid–base disturbances.²⁷ In addition to having a major role in H^+ secretion, H^+ -ATPase also has an essential role in cell volume regulation and maintenance of intracellular electronegativity, replacing the Na^+ - K^+ -ATPase that provides these functions in most other cell types.⁷³

A second means of H^+ secretion involves electroneutral, K^+ -dependent H^+ - K^+ -ATPase activity that is mediated by P-type H^+ - K^+ -ATPase proteins.¹⁵⁵ At least two H^+ - K^+ -ATPase α -isoforms are present. One, $HK\alpha_1$, is similar to the α -isoform involved in gastric acid secretion. The other, $HK\alpha_2$, is similar to the α -isoform in the colon. K^+ reabsorbed via apical H^+ - K^+ -ATPase can either recycle across the apical plasma membrane or exit the cell across the basolateral plasma membrane, and relative movement across the apical versus basolateral plasma membranes is regulated by dietary K^+ intake.⁴⁸⁴

A truncated isoform of the erythrocyte anion exchanger, termed kAE1, is present in the basolateral plasma membrane and mediates basolateral bicarbonate exit.⁹ Cl^- that enters the cell via basolateral Cl^-/HCO_3^- exchange exits via the KCl cotransporter, $KCC4$ ^{41,264}; a basolateral Cl^- channel, presumably $ClC-Kb$ in humans and $ClC-K2$ in rodents, also contributes to Cl^- recycling.²¹⁰

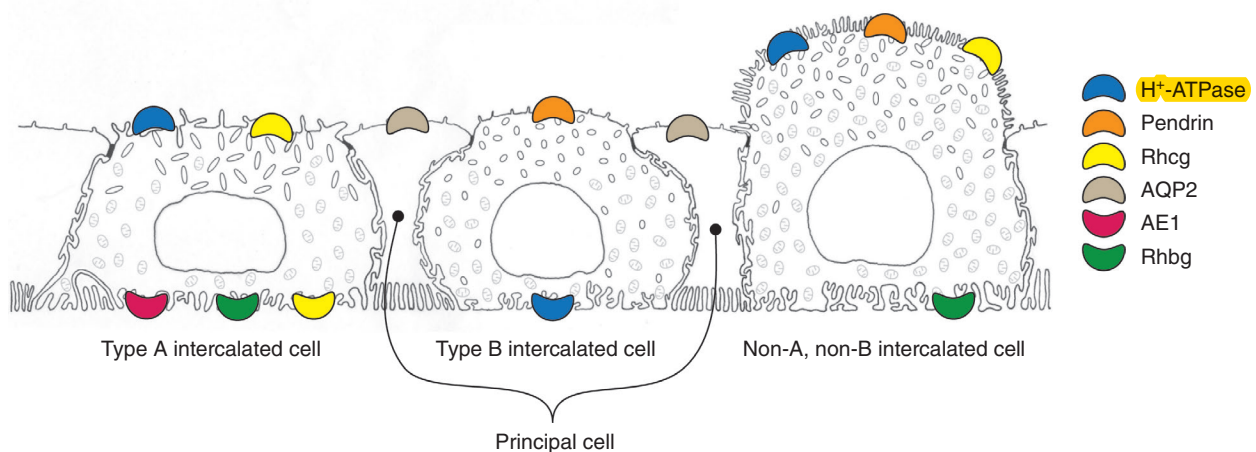
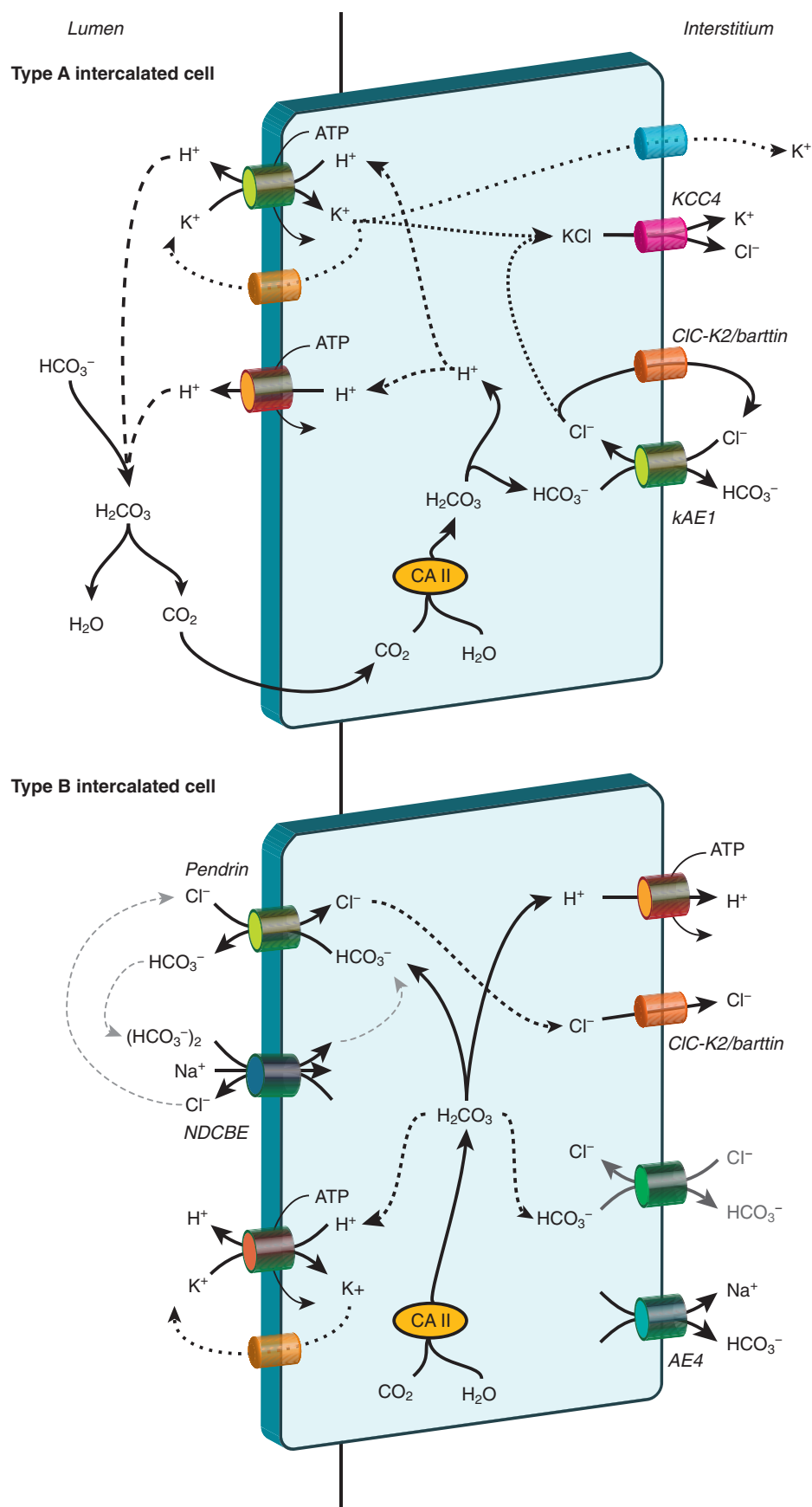


Fig. 9.3 Intercalated cell subtypes in the distal nephron and collecting duct. The late DCT, connecting segment, initial collecting tubule, CCD, OMCD, and IMCD have multiple distinct cell types. Three intercalated cell types can be distinguished based on ultrastructural features and differential expression in plasma membrane domains of several proteins involved in renal acid–base transport, including H^+ -ATPase, AE1, pendrin, Rhbg, and Rhcg. H^+ -ATPase is present in the apical portion of the type B intercalated cell, where it is found in cytoplasmic vesicles; ultrastructural analysis shows it is not in the apical plasma membrane. These specific intercalated cell subtypes occur at different frequencies specific to the various tubule segments. CCD, Collecting duct system; DCT, distal convoluted tubule; IMCD, inner medullary collecting duct; OMCD, outer medullary collecting duct.



Cytoplasmic CA II is abundant in type A intercalated cells and enables intracellular generation of H^+ for apical secretion and HCO_3^- for basolateral transport. In addition, membrane-associated carbonic anhydrases are present in the apical region (CA IV) and, at least in mouse and rabbit, in the basolateral region (CA XII) of intercalated cells.³¹⁶

Type B Intercalated Cell

The type B intercalated cell mediates a major role in HCO_3^- secretion and luminal Cl^- reabsorption. It contains basolateral H^+ -ATPase and an apical Cl^-/HCO_3^- exchanger, pendrin.¹⁰ H^+ -ATPase is also present in vesicles throughout the cell, but it is not present in the apical plasma membrane. Similar to the type A intercalated cell, H^+ -ATPase, rather than Na^+ - K^+ -ATPase, maintains intracellular electronegativity and prevents cell swelling.⁷³ Type B intercalated cells also express H^+ - K^+ -ATPase.^{421,473} In the rabbit and mouse an apical H^+ - K^+ -ATPase activity is present,^{250,456} whereas in the rat a basolateral H^+ - K^+ -ATPase activity may be present.¹³³ The type B cell also has cytoplasmic CA II, which facilitates intracellular H^+ and HCO_3^- production. Fig. 9.4 summarizes the proteins involved in type B intercalated cell acid-base transport.

There is functional evidence that pendrin-mediated apical Cl^-/HCO_3^- exchange works in concert with an apical Na^+ -dependent Cl^-/HCO_3^- exchanger, NDCBE (*Slc4a8*), to mediate net $NaCl$ absorption,²³⁷ and NDCBE protein and mRNA have been reported in renal cortical homogenates.^{237,468} However, in recent studies using single-cell RNA-seq NDCBE transcripts were virtually undetectable in type B intercalated cells.⁸⁰ Basolateral Na^+ and Cl^- exits are mediated by the basolateral $NaHCO_3$ cotransporter, AE4,⁷³ and a Cl^- channel, ClC -K2/Barttin, or ClC -Kb.¹⁷⁰

The type B intercalated cell also has the ability to secrete H^+ and reabsorb luminal HCO_3^- . As noted earlier, most studies indicate the type B intercalated cell has an apical H^+ - K^+ -ATPase activity that may mediate proton secretion, and functional studies have shown that CCD intercalated cells with apical Cl^-/HCO_3^- exchange activity (i.e., all type B intercalated cells) also have basolateral Cl^-/HCO_3^- exchange activity that is functionally distinct from the $kAE1$ activity that present in type A intercalated cells.⁴⁵⁹

The type B intercalated cell has several roles in acid-base and ion transport homeostasis. Genetic deletion of the apical Cl^-/HCO_3^- exchanger, pendrin, impairs HCO_3^- secretion, luminal Cl^- reabsorption, and, through a mechanism involving coordinated function with principal cell luminal Na^+ reabsorption.^{203,339,413} The type B intercalated cell may also contribute

to H^+ secretion and luminal HCO_3^- reabsorption. Ammonia, which is increased in metabolic acidosis and hypokalemia, increases type B intercalated cell apical H^+ - K^+ -ATPase and basolateral Cl^-/HCO_3^- exchange activity, which would result in increased net HCO_3^- reabsorption.¹²¹

Non-A, Non-B, or Type C Intercalated Cell

A third intercalated cell subtype, generally termed the non-A, non-B cell, is present in the CNT and ICT.^{201,391} This cell has several features that distinguish it from both the type A intercalated cell and the type B intercalated cell. These differences include the expression of both pendrin and H^+ -ATPase in the apical plasma membrane and in apical cytoplasmic vesicles, the absence of basolateral plasma membrane H^+ -ATPase and AE1, the presence of apical, but not basolateral Rhcg, and the absence of basolateral Rhbg (see Fig. 9.3). Thus, it differs significantly from both type A and type B intercalated cells. Studies of the developing kidney show that non-A, non-B cells and type B intercalated cells arise simultaneously, but from different foci.^{164,374} This cell type was termed “non-A, non-B cell” in early studies. However, its unique transporter expression, distribution, and developmental origin suggest this is a third distinct intercalated cell subtype.

Principal Cells

Principal cells have indirect and direct roles in acid secretion. Indirectly, principal cell-mediated Na^+ reabsorption leads to luminal electronegativity; this facilitates H^+ secretion by the electrogenic, and thus voltage-sensitive, H^+ -ATPase. In addition, principal cells have direct roles. Functional studies show that principal cells have apical H^+ secretory and basolateral Cl^-/HCO_3^- exchange activities,^{452,460} and they express H^+ -ATPase^{102,425} and both the $HK\alpha_1$ and $HK\alpha_2$ isoforms of H^+ - K^+ -ATPase.¹⁵⁵ In the mouse and rat kidney, principal cells in the OMCDi and initial IMCD express both carbonic anhydrase activity and CA II protein.^{101,205} Finally, the ammonia transporters, Rhcg and Rhbg, are both present in principal cells in the rat and mouse CCD and OMCD.¹⁹⁹

IMCD Cell

The IMCD cell is a distinct cell type and is the predominant cell present in the terminal IMCD. It exhibits carbonic anhydrase activity,²⁰⁵ both H^+ -ATPase and H^+ - K^+ -ATPase activity,^{155,435} and basolateral Cl^-/HCO_3^- exchange.⁴⁴⁹ In vitro microperfusion studies have demonstrated directly that the IMCD secretes H^+ and reabsorbs luminal HCO_3^- .⁴³⁷

Fig. 9.4 Bicarbonate transport by the type A and the type B intercalated cell. *Top panel* shows a model of acid-base transport by the type A intercalated cell. Two families of H^+ transporters, H^+ -ATPase and H^+ - K^+ -ATPase, are present in the apical plasma membrane. Secreted H^+ titrates luminal HCO_3^- to form H_2CO_3 , which dehydrates to water (H_2O) and carbon dioxide (CO_2). Luminal carbonic anhydrase activity, most likely mediated by CA IV, is variably present in the collecting duct (see text for details). Cytosolic H^+ and HCO_3^- are formed from CA II-accelerated hydration of CO_2 and rapid dissociation of H_2CO_3 . Cytosolic HCO_3^- exits across the basolateral plasma membrane via the anion exchanger, $kAE1$. Cl^- that enters via $kAE1$ recycles via a basolateral Cl^- channel. K^+ that enters via apical H^+ - K^+ -ATPase can either recycle via an apical, Ba^{2+} -sensitive K^+ channel or be reabsorbed via a basolateral Ba^{2+} -sensitive K^+ channel. A basolateral Na^+/H^+ exchanger is present but does not contribute to bicarbonate reabsorption and is not shown. *Bottom panel* shows a model of acid-base transport by the type B intercalated cell. Apical pendrin is the primary mechanism of bicarbonate secretion. Chloride enters the cell via pendrin and exits across a basolateral chloride channel, ClC -K2/barttin. Basolateral H^+ -ATPase extrudes protons into the peritubular compartment. Cytoplasmic bicarbonate and protons are produced from CO_2 and water in a CA II-catalyzed reaction. In addition, an apical H^+ - K^+ -ATPase in series with a basolateral Cl^-/HCO_3^- exchange activity is present and may contribute to bicarbonate reabsorption by the type B intercalated cell. Apical NDCBE is present and mediates $Na^+-(HCO_3^-)_2$ exchange for Cl^- . When coupled with pendrin this can enable coupled Na^+-Cl^- reabsorption. CA II, Carbonic anhydrase II.

FUNCTIONAL ROLE OF DIFFERENT COLLECTING DUCT SEGMENTS

CNT-ICT

Relatively little information is available on the functional role of the CNT and ICT in acid–base homeostasis. Morphologic and immunolocalization studies suggest that the CNT and ICT contain type A and type B intercalated cell types and non-A, non-B cells.^{201,391,419} Under basal conditions, the CNT, at least in the rabbit, secretes bicarbonate through a Cl^- , carbonic anhydrase–, and H^+ -ATPase–dependent mechanism⁴⁰⁴; this likely involves apical pendrin, cytosolic CA II, and basolateral H^+ -ATPase.

CCD

Unlike the OMCD and IMCD, which can secrete only acid (i.e., reabsorb bicarbonate), the CCD both reabsorbs and secretes bicarbonate. The basal direction of bicarbonate transport varies among species, but both bicarbonate absorption and secretion can be induced in response to systemic acid or alkali loading.^{18,245,261} The ability to secrete bicarbonate, which is not found in the OMCD or IMCD, correlates with the presence of type B intercalated cells in the CCD, but not in the OMCD or IMCD. Mineralocorticoids stimulate CCD bicarbonate secretion, likely related to generation of metabolic alkalosis and to stimulation of pendrin expression^{128,413}; however, mineralocorticoid receptors are present in type B intercalated cells,^{288,363} and thus direct stimulation of type B cell ion transport by mineralocorticoids is possible.

OMCD

The OMCD is responsible for approximately 40%–50% of the net acid secretion that occurs in the collecting duct. Both intercalated cells and principal cells contribute to acid secretion, although intercalated cells are believed to be the primary cell responsible for OMCD acid secretion.^{452,460}

IMCD

The IMCD secretes H^+ and reabsorbs luminal bicarbonate.⁴³³ However, the number of type A intercalated cells is substantially less than in other collecting duct segments. In the rat they account for only 10% of cells in IMCD1⁸⁶ and in all species examined the prevalence diminishes distally such that almost no intercalated cells exist in the distal portion of IMCD (IMCD3). Nonetheless, bicarbonate reabsorption occurs in the terminal IMCD and basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchange is present in cultured IMCD cells.⁴⁴⁹ H^+ secretion is partly mediated by H^+ - K^+ -ATPase.⁴³⁹ In rats fed a potassium-deficient diet, H^+ - K^+ -ATPase activities were upregulated,⁴³⁵ but H^+ - K^+ -ATPase accounted for only ~50% of bicarbonate reabsorption in the IMCD, indicating that other mechanisms of luminal acidification also contribute, likely including H^+ -ATPase. The IMCD expresses CA IV and luminal, cytoplasmic, and lateral membrane–associated carbonic anhydrase activity has been reported.^{205,432}

PROTEINS INVOLVED IN COLLECTING DUCT H^+ /BICARBONATE TRANSPORT

Collecting duct H^+ and HCO_3^- transport involves the coordinated activity of multiple transporters in conjunction with specific carbonic anhydrase isoforms. Later in this chapter we review the specific proteins involved.

H^+ -ATPase

Electrogenic apical H^+ secretion in acid-secreting intercalated cells and basolateral proton transport by bicarbonate-secreting intercalated cells is mediated by the vacuolar H^+ -ATPase. Intercalated cells in the OMCD and initial IMCD, type A intercalated cells in the CCD, and non-A, non-B intercalated cells in the CNT and CCD, contain H^+ -ATPase in the apical plasma membrane and in an apical cytoplasmic vesicle pool, and redistribution between the cytoplasmic vesicle pool and the apical plasma membrane is a major mechanism regulating H^+ secretion. H^+ -ATPase is also present in the apical region of principal cells and CNT cells, but expression is much less than in intercalated cells. In Type B intercalated cells, H^+ -ATPase is present in the basolateral plasma membrane and in sub-apical vesicles, but not in the apical plasma membrane. The role of apical H^+ -ATPase in nonintercalated cells has not been clearly defined; it may be involved in endosomal trafficking and fusion,¹⁸⁴ and in the OMCDi it can also mediate apical H^+ secretion.⁴⁵²

Vacuolar H^+ -ATPase is an assembly of multiple subunits that form two main domains, the V_1 domain, which is extramembranous and hydrolyzes ATP, and the V_0 domain, which is a transmembranous portion and transports protons. The V_0 domain is composed of six subunits; the V_1 domain is composed of eight subunits and is linked to the V_0 domain via a stalk region composed of subunits from both V_0 and V_1 . Distinct isoforms and splice variants have been identified for many of these H^+ -ATPase subunits and their cell-specific distribution may contribute to cell-specific regulation of proton and bicarbonate transport.

Genetic defects in several H^+ -ATPase subunits have been shown to cause distal RTA (dRTA), also known as Type I RTA, in humans. Defects in the B1 subunit of the hydrolytic V_1 domain resulting from ATP6V1B1 gene mutations can produce early-onset hearing loss in combination with autosomal recessive, severe dRTA.^{194,195,382} Mice with B1 subunit deletion have incomplete distal RTA.¹¹⁶ In these mice, the B2 subunit appears to substitute partially for the B1 subunit, enabling partial compensation.³⁰¹ Mutations in the $\alpha 4$ subunit (ATP6V0A4) in the H^+ -translocating V_0 domain also produce recessive, severe early onset of dRTA, with variable onset of hearing loss.^{371,382,411}

Normal H^+ -ATPase function appears to involve coexpression of the Atp6ap2/(pro)renin receptor. This is a type 1 transmembrane protein and an accessory subunit of H^+ -ATPase and may also function in the renin-angiotensin system. Its deletion results in decreased H^+ -ATPase activity and impaired renal acid–base homeostasis.³⁹⁹ Although Atp6ap2 is a cell surface protein capable of binding and nonproteolytically activating prorenin, prorenin does not acutely regulate H^+ -ATPase activity.⁹⁶ Atp6ap2 also appears to regulate NKCC2 and AQP2 expression, and this may occur through the autophagosomal substrate p62.³⁹⁹

H^+ - K^+ -ATPase

The second mechanism of collecting duct H^+ secretion involves electroneutral H^+ - K^+ exchange.¹⁵⁵ The active protein is a heterodimer composed of α - and β -subunits. The α -subunit is an integral membrane protein with multiple membrane spanning domains and contains the catalytic portion of the enzyme. Two α -subunit isoforms have been identified. HK α_1 , also termed the gastric isoform, was identified originally in

the stomach. HK α_1 forms heterodimers with its specific β -subunit, HK β . The β -subunit has only a single membrane-spanning region and is necessary for targeting of the α -subunit to the plasma membrane and for transport function.¹⁵⁵ HK α_2 was identified originally in the colon and is sometimes referred to as the colonic isoform. Three splice variants of HK α_2 have been identified in the kidney. HK α_2 forms heterodimers with the β_1 -subunit of Na⁺-K⁺-ATPase.

HK α_1 , HK α_2 , and HK β are expressed throughout the collecting duct, with greater expression in intercalated cells than in principal cells.^{4,6,64} Functional studies suggest that both HK α_1 and HK α_2 are present in type A as well as type B intercalated cells.^{250,265,456} However, immunohistochemistry studies have yielded variable results with respect to the precise cellular distribution of the HK α_1 and HK α_2 isoforms. HK α_1 immunoreactivity was found in both AE1-positive (type A) and AE1-negative intercalated cells in both rat and rabbit collecting ducts,⁴⁷³ but in human kidneys, diffuse HK α_1 immunoreactivity was present in both intercalated and principal cells.²¹⁵ HK α_2 immunoreactivity was consistently apical, but in different cell types in different studies. It was found exclusively in the CNT cell in rabbits in one study¹¹⁴ and exclusively in the OMCD principal cell in rats in another.³⁴³ A third study found the splice variant, HK α_{2c} , in intercalated cells, principal cells, and CNT cells from the CNT through the initial IMCD in rabbit kidney.⁴²¹ In situ hybridization studies have shown both intercalated cell and principal cell expression of HK α_1 , HK α_2 , and HK β mRNA in the rat kidney, although principal cell signal was less intense than intercalated cell signal.^{5,64}

Multiple physiologic conditions alter H⁺-K⁺-ATPase expression and activity. Metabolic acidosis increases H⁺-K⁺-ATPase activity in the CCD and HK α_1 and HK α_2 mRNA expression in the OMCD, suggesting that H⁺-K⁺-ATPase contributes to H⁺ secretion.¹⁵⁵ Specific studies have identified apical, but not basolateral, H⁺-K⁺-ATPase activity in both types A and B intercalated cells in mouse and rabbit kidneys.^{265,456,483} Extracellular ammonia, which increases with both metabolic acidosis and hypokalemia, enhances apical H⁺-K⁺-ATPase-mediated H⁺ secretion in both type A and type B intercalated cells in the CCD.^{121,123}

Pendrin (SLC26A4)

Pendrin is an electroneutral Cl[−]/HCO₃[−] exchanger present in the kidney exclusively in type B and non-A, non-B intercalated cells. It is found in the apical plasma membrane and in apical cytoplasmic vesicles in type B and non-A, non-B intercalated cells in the CNT, ICT, and CCD. Under basal conditions, pendrin is predominantly expressed in the apical plasma membrane in non-A, non-B intercalated cells and in subapical cytoplasmic vesicles in type B intercalated cells, and redistribution between these two subcellular sites is an important regulatory mechanism.⁴⁴¹ Pendrin is regulated by AngII, nitric oxide, and cAMP.^{393,414} In addition to bicarbonate secretion, pendrin also mediates an important role in extracellular fluid volume and blood pressure regulation. This appears to involve roles in both transcellular Cl[−] reabsorption and, through luminal alkalization due to HCO₃[−] secretion, activation of the principal cell epithelial Na⁺ transporter, ENaC^{108,436} and involving the Na⁺-dependent, chloride–bicarbonate exchanger (NDCBE, SLC4A8).²³⁷

Carbonic Anhydrase

Three carbonic anhydrase isoforms, CA II, CA IV, and CA XII, are present in the collecting duct. CA II is cytosolic in proximal tubule cells, discussed earlier, in intercalated cells, and in principal cells in the collecting ducts of mice.⁴²⁰ CA II is present in all intercalated cell types, but expression is generally greater in the type A than in type B intercalated cells.

CA IV is an extracellular, membrane-associated carbonic anhydrase tethered to the membrane through a glycosylphosphatidylinositol lipid (GPI) anchoring protein. It is expressed apically in the majority of cells in rabbit OMCD and IMCD and in type A intercalated cells in the CCD.³⁵⁶ In the OMCDi, luminal carbonic anhydrase inhibition decreases bicarbonate absorption, suggesting an important role for CA IV in acid–base homeostasis.⁴⁰³

Carbonic anhydrase XII (CA XII) is another extracellular, membrane-associated carbonic anhydrase found in the collecting duct.³¹⁶ In contrast to CA IV, CA XII is an integral membrane protein with a single transmembrane spanning region.³¹⁶ Basolateral CA XII immunoreactivity has been reported in principal cells in the human kidney, and in the mouse, basolateral CA XII immunoreactivity is found in type A intercalated cells in the CCD and OMCD.^{316,319}

kAE1

The major basolateral anion exchanger in type A intercalated cells is kAE1, a truncated form of the erythrocyte anion exchanger AE1. In the human, rat, and mouse kidneys, kAE1 is expressed almost entirely in the basolateral plasma membrane. In the rabbit kidney under basal conditions, kAE1 is present in intracytoplasmic multivesicular bodies, as well as in the basolateral plasma membrane; metabolic acidosis decreases intracellular kAE1 and increases basolateral kAE1, suggesting regulated trafficking contributes to bicarbonate reabsorption.⁴¹⁶

Several mutations in AE1 cause human autosomal dominant and autosomal recessive dRTA. Autosomal dominant dRTA can be caused by a trafficking defect leading either to mistargeting to the apical plasma membrane or failure of plasma membrane insertion.^{324,362} Autosomal recessive dRTA due to defective AE1 is commonly due to mutations that lead to intracellular protein retention.³⁸⁸

KCC4

KCC4 is a member of the SLC12 family of solute transporters and mediates electroneutral, coupled transport of K⁺ and Cl[−]. Basolateral KCC4 expression has been shown in the proximal convoluted tubule (PCT), TAL, DCT, CNT, and type A intercalated cells.^{41,412} In the type A intercalated cell, KCC4 likely contributes to basolateral Cl[−] recycling. Metabolic acidosis increases KCC4 expression in type A intercalated cells in the OMCD, suggesting a role in the response to metabolic acidosis²⁶⁴ by facilitating basolateral Cl[−]/HCO₃[−] exchange, and KCC4 deletion causes development of distal RTA,⁴¹ suggesting KCC4 is necessary for both basal and acidosis-stimulated acid–base homeostasis.

Cl[−] Channel

Cl[−] entry via basolateral kAE1 recycles across the basolateral plasma membrane. In addition to KCC4, the Cl[−] channel, ClC-K2, is present in the basolateral plasma membrane of type A intercalated cells and likely contributes to this recycling.²⁹⁶

Other Anion Exchangers

Several other anion transporters, including anion exchangers and sodium bicarbonate cotransporters (NBCs), are present in the collecting duct, but their roles in acid–base homeostasis are less completely understood. AE2 is expressed in collecting ducts, particularly in the basolateral plasma membrane of IMCD cells.¹²⁵ Another $\text{Cl}^-/\text{HCO}_3^-$ exchanger, Slc26a7, is found in the basolateral plasma membrane of OMCD intercalated cells.³⁰⁷ Slc26a7 mRNA and protein expression increases with acid loading, suggesting it may contribute to regulated bicarbonate reabsorption.³⁸⁴ AE4 (Slc4a9) is present in the collecting duct, but both its location and function are in question. Although originally thought to be an anion exchanger, other evidence suggests AE4 functions in type B intercalated cells as a NaHCO_3 cotransporter.⁷³ Studies of the rabbit kidney have variously reported that AE4 immunoreactivity is exclusively apical in type B intercalated cells,⁴⁰¹ apical and lateral in type A intercalated cells,²⁰⁹ and exclusively lateral in type B intercalated cells.³¹⁸ In both rat and mouse kidneys, only basolateral AE4 protein expression has been detected in both type A^{80,209} and type B intercalated cells.^{73,80,209} In the mouse, both mRNA and basolateral protein expression are significantly stronger in type B than in type A intercalated cells.⁸⁰

Sodium Bicarbonate Cotransporters

Several NBCs are expressed in the collecting duct. NBC3 (Slc4a7) is found in the apical region of OMCD intercalated cells and type A intercalated cells in the CCD and in the basolateral region of type B intercalated cells.^{221,320} It appears to contribute to intracellular pH regulation, but not to transepithelial bicarbonate transport.⁴⁸² NBCn1, another SLC4A7 gene product, is an electroneutral NBC, and basolateral expression is found in the terminal IMCD and in OMCD intercalated cells.³¹¹ Finally, recent evidence indicates AE4 (SLC4A9), discussed earlier, functions as a $\text{Na}^+\text{-HCO}_3^-$ cotransporter.⁷³

NBCe2 (SLC4A5)

The electrogenic, $\text{Na}^+\text{-HCO}_3^-$ cotransporter, isoform 2 (NBCe2, SLC4A5) appears to contribute to acid–base homeostasis, but its specific cellular expression remains unclear. An initial study suggested there was only apical expression in collecting duct intercalated cells in the CCD and OMCD,⁹⁵ whereas a different study identified NBCe2 in microdissected mouse CNT segments.⁴⁶⁶ Genetic deletion in one study induced metabolic acidosis,¹⁵⁰ whereas in a second study, NBCe2 deletion did not alter basal acid–base homeostasis, but did impair the ability to respond to exogenous acid loading.⁴⁶⁶ The mechanism through which NBCe2 contributes to acid–base homeostasis, given the evidence for its apical localization, is unclear. There is evidence that NBCe2 deletion leads to increased pendrin expression, which could, by increasing HCO_3^- secretion, contribute to the acid–base phenotype observed.⁹⁵ NBCe2 deletion is also associated with increased expression of the β_1 -subunit of H^+ -ATPase, which may serve as an adaptive response that minimizes the extent of the acid–base disturbance.⁴⁶⁶ However, the possibility of additional mechanisms through which NBCe2 contributes to acid–base homeostasis cannot be excluded at present.

Other studies suggest NBCe2 is present in the proximal tubule, not the distal nephron. In a study examining the

human kidney under basal conditions, NBCe2 mRNA was found in the proximal tubule by in situ hybridization, and protein expression was observed in subcellular fractionation enriched for apical brush-border proteins.¹³⁵

REGULATION OF COLLECTING DUCT ACID–BASE TRANSPORT

The collecting duct is the final site controlling renal acid–base regulation. It responds quickly to physiologic conditions to increase acid or bicarbonate excretion as needed to maintain systemic acid–base homeostasis.

ACIDOSIS

The collecting duct response to metabolic acidosis includes adaptations in all segments of the collecting duct and the CNT. Increased acid secretion in the collecting duct during acidosis is mediated primarily by H^+ -ATPase. Both metabolic and respiratory acidosis increase apical plasma membrane H^+ -ATPase expression and activity in acid-secreting collecting duct intercalated cells. Redistribution of H^+ -ATPase from a subapical vesicle pool to the apical plasma membrane is the primary means of activation of proton secretion, and involves vesicular trafficking that requires soluble NSF-attachment protein receptor (SNARE) proteins and Rab GTPases.^{102,425} In most models of metabolic acidosis, total renal H^+ -ATPase mRNA and protein expression do not change,^{379,425} but a study examining OMCD segments from acid-loaded mice found increased mRNA expression of several H^+ -ATPase subunits, including the B1 and a4 subunits.⁸²

During chronic metabolic acidosis, AE1 mRNA and AE1 protein expression in the basolateral plasma membrane in OMCD and CCD type A intercalated cells is increased.^{181,341} In rats and mice, AE1 is present in the basolateral plasma membrane under basal conditions, and the subcellular distribution does not change with metabolic acidosis.^{181,341} In rabbits fed a normal diet, AE1 is in both intracellular multivesicular bodies and the basolateral plasma membrane in type A intercalated cells. Metabolic acidosis increases the basolateral plasma membrane boundary length and the amount of AE1 immunoreactivity in the basolateral plasma membrane and reduces intracellular AE1.⁴¹⁶

During metabolic acidosis, both net HCO_3^- secretion and type B intercalated cell-mediated unidirectional HCO_3^- secretion decrease. This is associated with decreased pendrin expression in type B and non-A, non-B cells as well as decreased apical $\text{Cl}^-/\text{HCO}_3^-$ exchange activity in type B intercalated cells in the CCD.^{124,426,475} Reduced bicarbonate secretion by B cells during acid loading thus contributes to increased net bicarbonate reabsorption.

Carbonic anhydrase activity and the expression of CA II and CA IV in the collecting duct are increased by metabolic acidosis.³¹⁶ CA IV expression is upregulated in the OMCD, whereas CA II expression is upregulated in the CNT, CCD, and OMCD.

The collecting duct response to respiratory acidosis appears to be similar to that of metabolic acidosis. Respiratory acidosis stimulates structural changes in OMCD and CCD type A intercalated cells consistent with translocation of H^+ -ATPase-bearing membrane from the apical vesicle pool to the apical plasma membrane.⁴¹⁸ Respiratory acidosis also stimulates N-ethylmaleimide-sensitive ATPase activity, a measure of H^+ -ATPase activity^{106,425} and bicarbonate reabsorption in isolated

CCDs,²⁶² consistent with activation of H⁺-ATPase mediated proton secretion. In addition, chronic respiratory acidosis increases kAE1 mRNA.⁹⁴ Pendrin expression decreases during respiratory acidosis,⁹⁸ which likely mediates decreased bicarbonate secretion.

ALKALOSIS

Metabolic alkalosis induces coordinated changes in acid–base transport throughout the collecting duct. In the OMCD of bicarbonate-loaded animals, bicarbonate reabsorption is decreased compared with control animals²⁴⁵ and in the IMCD, bicarbonate loading abolishes acid secretion.³² In the CCD, bicarbonate loading in animals produces net bicarbonate secretion.²⁶¹ However, no studies have shown the development of HCO₃[−] secretion by the OMCD or the IMCD in response to metabolic alkalosis, and this correlates with the lack of pendrin-expressing type B and non-A, non-B intercalated cells in these segments.

The cellular response to alkalosis in OMCD and CCD type A cells entails essentially the reverse of processes that occur to stimulate acid secretion. H⁺-ATPase is redistributed from the apical plasma membrane into the apical vesicle pool, and basolateral AE1 immunoreactivity decreases.^{26,341,417} Depending on the animal model, alkalosis increases pendrin expression and its apical distribution in type B and non-A, non-B intercalated cells and increases pendrin-mediated CCD bicarbonate secretion.^{124,426} However, pendrin expression, subcellular location, and functional activity are regulated by other factors independent of acid–base status, including pregnancy, aldosterone, AngII, activation of AT1a and AT1b receptors, nitric oxide, and cAMP,* in addition to chloride balance and luminal chloride delivery.^{321,415}

HORMONAL REGULATION OF COLLECTING DUCT ACID–BASE TRANSPORT

In addition to extracellular pH, multiple other factors regulate collecting duct acid–base transport. Importantly, in vivo acid–base changes cause greater adaptations than equivalent in vitro changes, suggesting that in vivo regulatory mechanisms mediate a critical role in the response to acid–base disturbances.¹³⁴ Several hormones and receptors regulate bicarbonate transport in the collecting duct, particularly aldosterone and its analogs, and AngII.

Aldosterone is an important regulator of collecting duct bicarbonate transport.³⁸¹ Both in vivo and in vitro mineralocorticoids increase OMCD bicarbonate reabsorption.³⁸¹ This involves, at least when studied in vitro, increased H⁺-ATPase activity and apical translocation in OMCD intercalated cells, stimulated through a non-genomic pathway not inhibited by mineralocorticoid receptor blockade.⁴⁷⁴ Mineralocorticoids also increase CCD bicarbonate secretion; this is dependent on luminal chloride, mediated by pendrin, and involves increased pendrin mRNA and protein expression and pendrin redistribution from cytoplasmic vesicles to the apical plasma membrane in type B intercalated cells.^{339,413} Likely because of parallel stimulation of both acid and bicarbonate secretion, mineralocorticoid therapy usually has modest effects on systemic acid–base homeostasis.

AngII exerts effects on the proximal tubule, TAL, DCT and collecting ducts. The collecting duct expresses apical

AT1 (AT1a) receptors in both principal cells and intercalated cells.³³⁸ In mouse OMCD and CCD, AngII in vitro increases H⁺-ATPase activity in acid-secreting intercalated cells by trafficking H⁺-ATPase to the apical plasma membrane.^{304,338} In mouse OMCD, AngII stimulates H⁺-ATPase activity through a G-protein–coupled phosphokinase C pathway.³³⁸ However, in other studies, in vivo and in vitro AngII decreased bicarbonate reabsorption in rat OMCD and in vitro AngII decreased H⁺-ATPase activity via AT1 receptors^{394,440}; this apparent discrepancy has not been resolved. AngII also increases pendrin-dependent Cl[−] absorption,³⁰² apical Cl[−]/HCO₃[−] exchange in type B intercalated cells,⁴⁵⁷ and pendrin protein expression in the apical plasma membrane in non-A, non-B intercalated cells in the CNT, an effect mediated through activation of the angiotensin type 1a receptor (Agtr1a).⁴¹⁴

Endothelin has important effects on collecting duct acid–base transport that are mediated partly by nitric oxide. Dietary protein intake stimulates urinary acidification through a process involving H⁺-ATPase activation, mediated by endothelin and nitric oxide.⁴⁶⁷ Endothelin-1 (ET-1) is synthesized by the collecting duct,^{364,383} and endothelin receptors A and B (ET-A and ET-B) are present in the collecting duct.²¹² ET-B activation regulates both type A and type B intercalated cell responses to metabolic acidosis.⁴⁰⁵

The CaSR is apical in IMCD cells and in type A intercalated cells³³¹ and mediates luminal Ca⁺²-stimulation of H⁺-ATPase.³³⁰ Luminal acidification stimulated by this pathway may inhibit calcium precipitation and minimize development of nephrolithiasis.³³⁰

Activation of the vasopressin type 1A (V1a) receptor is an additional regulatory mechanism. The V1a receptor is expressed in the medullary TAL (mTAL) and throughout the collecting duct,^{70,481} with expression in both intercalated cells and principal cells in the CCD and only in intercalated cells in the OMCD.⁷⁰ Metabolic acidosis increases V1a receptor expression in the mTAL and the OMCD in the inner stripe.^{70,389} Genetic deficiency of the V1a receptor causes development of type IV RTA and diminishes mineralocorticoid stimulation of H⁺-K⁺-ATPase and Rhcg.¹⁸⁷

Several other hormones and drugs also alter collecting duct acid–base transport. Kallikrein inhibits bicarbonate secretion.²⁵³ Calcitonin stimulates H⁺-ATPase–dependent bicarbonate reabsorption in the rabbit CCD.³⁶⁵ Isoproterenol stimulates bicarbonate secretion by type B intercalated cells.³⁵¹

PARACRINE REGULATION

Several compounds produced and/or transported in the proximal tubule and TAL have downstream effects that regulate collecting duct acid–base transport. Presumably, this enables these segments, which exist in an area with very high blood flow and thus rapid exposure to changes in systemic acid–base and potassium, to regulate transport in collecting duct segments in the outer medulla and inner medulla, sites of low blood flow and thus reduced exposure to changes in systemic acid–base and potassium homeostasis. The paracrine molecules most extensively studied are ammonia and alpha-ketoglutarate.

Ammonia, discussed in detail later regarding its role in net acid excretion, also appears to function as an intrarenal, paracrine signaling molecule that regulates collecting duct transport.⁴⁵¹ It is produced primarily in the proximal tubule and undergoes regulated transport in both the proximal

*References 302, 303, 393, 413, 414, 468.

tubule and the TAL in response to both acid loading and hypokalemia. In addition to its roles in bicarbonate generation, ammonia stimulates CCD bicarbonate reabsorption in a concentration-dependent fashion.¹²³ Ammonia stimulates type A intercalated cell acid secretion and inhibits type B intercalated cell bicarbonate secretion.^{121,123} Its stimulation of proton secretion involves stimulation of $H^+-K^+-ATPase$, not $H^+-ATPase$, activity.^{122,123}

The Krebs cycle intermediate, 2-oxoglutarate (alpha-ketoglutarate), may have an important role in acid–base homeostasis. Changes in acid–base loading change the net direction of transport in the proximal tubule and the loop of Henle from reabsorption, seen with acid loading, to net secretion, seen with alkali loading.^{79,115,395} In the CNT and CCD, luminal 2-oxoglutarate enhances net bicarbonate and sodium chloride reabsorption, acting through its receptor, Oxgr1, in type B and non-A, non-B intercalated cells.³⁹⁵ Thus, 2-oxoglutarate can function as a paracrine mediator enabling functional coordination of the proximal tubule and the TAL with the collecting duct.

CELLULAR ADAPTATIONS TO ACID–BASE PERTURBATIONS

In addition to changes in the abundance and subcellular distribution of membrane transporters, adaptive responses to some physiologic disturbances may involve changes in the numbers of intercalated cells. Several studies have shown that chronic metabolic acidosis and chronic hypokalemia increase intercalated cell numbers in medullary collecting ducts,* whereas others find no change in intercalated cell number in these conditions.^{168,181,419} Chronic administration of lithium and acetazolamide also increases intercalated cell numbers in the OMCD.^{23,85,398}

Increases in intercalated cell numbers could result from intercalated cell proliferation or from principal cell proliferation followed by conversion into intercalated cells. Studies using proliferation markers show that metabolic acidosis, hypokalemia, and lithium administration are each associated with increased proliferation of collecting duct cells,^{84,297,465} some showing increased proliferation in type A intercalated cells,^{410,465} and others showing the proliferating cells are principal cells.^{84,202,297,398} The latter studies suggested that principal cells and OMCD intercalated cells may interconvert based on observations of rare cells with immunohistochemical and ultrastructural characteristics of both cell types^{84,297,398} and through genetic studies that irreversibly identify principal cells as cells that express genes under the control of the AQP2 promoter.²⁰² The histone H3 K79 methyltransferase, Dot1L, may be involved in preventing transformation of principal cells into intercalated cells; Dot1L deletion decreases the number of principal cells and increases the number of collecting duct intercalated cells.⁴⁷⁹ Other studies show that principal cells respond to acid by producing the cytokine SDF1, also known as CXCL12, which then acts on adjacent intercalated cells via its receptor, CXCR4.³⁵⁵ SDF1 is transcriptionally regulated and is a target of the hypoxia-sensing transcription factor HIF1 α in principal cells.³⁵⁵

With respect to the CCD, some studies suggested there may be interconversion of type A and type B intercalated cells. An early paper examining the rabbit isolated perfused

CCD equated apical endocytosis with alpha (type A) intercalated cells and used apical peanut lectin binding as a marker of beta (type B) intercalated cells. Chronic NH_4Cl loading in vivo increased the number of intercalated cells exhibiting apical endocytosis in microperfused CCDs and decreased the number of cells that bound peanut lectin; the interpretation was that intercalated cell subtypes in the CCD could interconvert, with the type B intercalated cells reversing polarity to meet the physiologic demand for increased acid secretion.³⁵³ Subsequently, some studies reported that acidosis, lithium administration, and carbonic anhydrase inhibition each alter the relative numbers of intercalated cells identifiable as type A or type B, although none have shown cells in native tissue with either apical AE1, basolateral pendrin, or coexpression of these two transporters.^{23,85,124,317,465} Other studies of acid–base disturbances find regulation of the abundance and distribution of transport proteins specific to the A and B intercalated cell types and changes in cell morphology, but no change in the relative or absolute numbers of specific intercalated cell subtypes.^{27,341,418,419} The explanation for these different findings could include differences in the experimental models, species examined, sensitivity and specificity of intercalated cell identification, and cell quantitation methods.

In vitro studies have implicated the extracellular matrix protein, hensin, and the prolyl isomerase activity of cyclophilin in the process of intercalated cell remodeling.^{354,387} In mice with intercalated cell-specific hensin deletion there is development of a distal RTA, lack of type A intercalated cells, and an increased number of type B intercalated cells.¹²⁷ Hensin's effects on type A intercalated cell development appears to require the activation of beta-1 integrin.¹²⁷

BICARBONATE GENERATION

Acid–base homeostasis requires not only reabsorption of filtered bicarbonate, but also the generation of new bicarbonate to replace the bicarbonate used for buffering of endogenous and exogenous fixed acids. There are two major components of bicarbonate generation, titratable acid excretion and ammonia excretion. In addition, organic anion excretion is biologically important. Organic anions can be metabolized to form HCO_3^- ; accordingly, their excretion is physiologically equivalent to bicarbonate excretion.

TITRATABLE ACID EXCRETION

Titrateable acids are urinary solutes that buffer secreted protons, enabling H^+ excretion without substantial changes in urine pH. Titratable acid excretion constitutes ~40% of net acid excretion under basal conditions. Metabolic acidosis increases titratable acid excretion by as much as 50% above baseline^{159,347} (Fig. 9.5).

Multiple buffers contribute to titratable acid excretion. An ideal urinary buffer has a pK_a lower than systemic pH, so that the majority of the filtered component is in the base form, and a pK_a higher than urine pH, so that the majority of the urinary form is in the acid form. Phosphate is the predominant titratable acid and typically accounts for more than 50% of total titratable acid.^{159,478} Citrate and creatinine also contribute to titratable acid excretion, but to a lesser

*References 23, 85, 297, 398, 410, and 465.

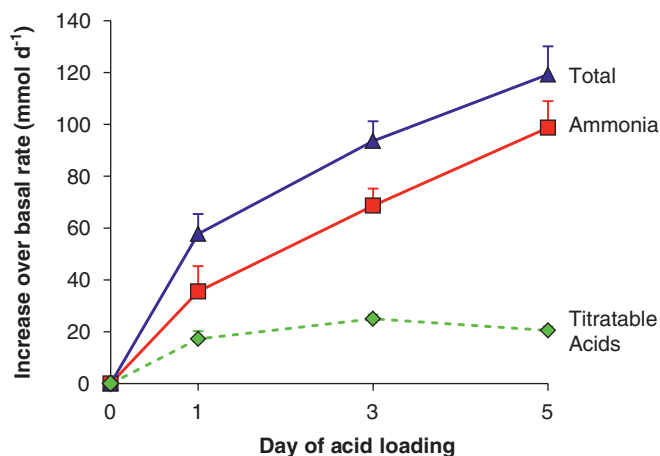


Fig. 9.5 Relative contribution of titratable acid and ammonia excretion in the response to metabolic acidosis. Normal human volunteers were acid-loaded with ~2 mmol/kg of ammonium chloride and changes in urinary ammonia and titratable acid excretion were quantified. Data recalculated from Elkinton et al.¹⁰⁹

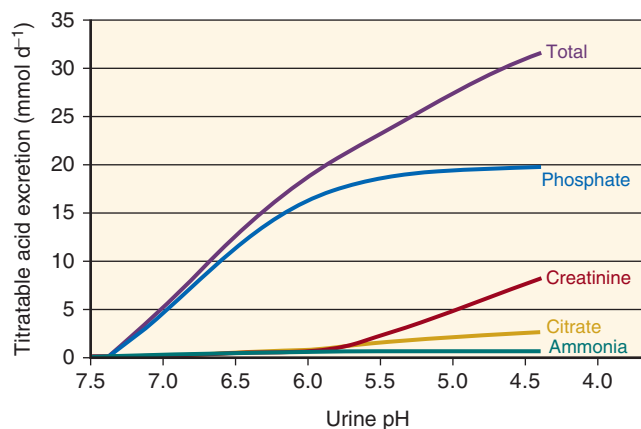


Fig. 9.6 Relative contribution of various urinary buffers to titratable acid excretion. Ability of various urinary buffers to contribute to titratable acid excretion depends on the amount excreted in the urine, their pK_a , and final urine pH. Figure shows titratable acid excretion accounted for by each of the four major urinary buffers, phosphate, creatinine, citrate, and ammonia, at differing urine pH. Rates were calculated with daily excretion rate and pK_a , respectively, for phosphate, 25 mmol/d and 6.8; creatinine, 11 mmol/d and 4.9; citrate, 3 mmol/d and 5.6; and ammonia, 40 mmol/d and 9.15.

extent. Although ammonia is frequently termed a urinary buffer, because of its high pK_a it does not contribute substantially to titratable acid excretion. The role of ammonia in new bicarbonate generation is considered separately later in the chapter. Fig. 9.6 shows the relative contributions of major urinary buffers to titratable acid excretion and shows the effect of changes in urine pH after taking into account the amount excreted under normal conditions and the pK_a of each buffer.

PHOSPHATE AS A TITRATABLE ACID

Titratable acid excretion in the form of phosphate is the amount of HPO_4^{2-} that is filtered, not reabsorbed, and that

buffers secreted H^+ , forming $H_2PO_4^-$. Phosphate exists, under physiologically relevant conditions, in equilibrium between two forms: $H_2PO_4^-$ and HPO_4^{2-} .² The relative amount of these two forms is given by

$$10^{pH-6.8} = \frac{[HPO_4^{2-}]}{[H_2PO_4^-]}.$$

The amount of $H_2PO_4^-$ in the urine ($H_2PO_4^-$ Urine) at any given pH can be calculated as

$$H_2PO_4^- \text{ Urine} = \frac{U_{\text{Phos}} V}{10^{(pH_U - 6.8)} + 1},$$

where U_{Phos} is the urinary concentration of total phosphate. Filtered phosphate, at the typical serum pH of 7.4, is ~80% in the form of HPO_4^{2-} and 20% in the form of $H_2PO_4^-$. Thus, at any urine pH (pH_U), titratable acid excretion in the form of phosphate (TA_{Phos}) is given by the formula

$$TA_{\text{Phos}} = U_{\text{Phos}} V * \left(\frac{1}{10^{(pH_U - 6.8)} + 1} - 0.2 \right).$$

These considerations indicate that titratable acid excretion as phosphoric acid is determined by phosphate excretion and by the ability to lower urine pH. Phosphate excretion is determined by the difference between the filtered load of phosphate and tubular phosphate reabsorption. Regulation of renal tubular phosphate transport is a complex process and is discussed in detail elsewhere in this text. Here, we review only the factors that regulate this process in response to acid-base disorders.

The proximal tubule is the primary site of phosphate reabsorption and is where metabolic acidosis and other acid-base disorders regulate phosphate transport. Acid loading decreases proximal tubule phosphate reabsorption, leading to increased excretion. However, absolute changes in urinary phosphate excretion are usually rather modest, less than a twofold increase. The decrease involves decreased NaPi-IIa protein and mRNA expression and changes in its subcellular distribution.^{15,15,156} Acid loading alters NaPi-IIa expression even if the acid load is completely compensated and there are no changes in systemic pH, suggesting that factors that precede changes in systemic pH regulate this response.⁴²³ Metabolic acidosis also lowers luminal pH in the proximal tubule, which directly inhibits phosphate uptake.^{174,407} Finally, metabolic acidosis increases PTH release, which also inhibits phosphate reabsorption.

Other phosphate transporters besides NaPi-IIa, such as NaPi-IIc and Pit-2, are present in the proximal tubule apical plasma membrane. Whether NaPi-IIc changes with metabolic acidosis is unclear as some studies find decreased expression⁴²³ and others do not.²⁹² Pit-2 expression, although regulated by dietary phosphate availability, is not altered in metabolic acidosis in phosphate replete conditions, but does increase in response to metabolic acidosis in conditions of phosphate depletion.⁴²³

Acidosis-induced changes in phosphate excretion depend on systemic phosphate availability. In the presence of dietary phosphate restriction, basal phosphate excretion is reduced, and the increase in urinary phosphate excretion in response to metabolic acidosis is blunted.⁴²³ Similarly, changes in NaPi-IIa abundance are blunted.¹⁵ In contrast to NaPi-IIa, and to their response to metabolic acidosis in

phosphate-replete animals, NaPi-IIc and Pit-2 expression actually increase in phosphate-restricted animals exposed to metabolic acidosis.⁴²³

Increased renal phosphate excretion with metabolic acidosis is balanced by parallel increases in extrarenal phosphate transport. Metabolic acidosis increases small intestinal Na⁺-dependent phosphate transport, and this is associated with increased expression of NaPi-IIb.³⁷⁸ There is also increased phosphate release from bone in response to both acute and chronic metabolic acidosis.²³⁴ These extrarenal effects minimize the changes in systemic phosphate levels that could otherwise develop from the increased phosphate excretion.

OTHER URINARY BUFFERS

Creatinine, which is used typically to assess glomerular filtration, has a pK_a of ~4.9 and is excreted in sufficient amounts, ~11 mmol d⁻¹, that it can contribute to titratable acid excretion. This is particularly true in conditions when urinary pH is 5.5 or less.¹⁵⁹ Uric acid, although it can function as a buffer, is typically excreted in such small amounts, ~4 mmol d⁻¹, as to limit its role as a titratable acid. In ketoacidosis, β-hydroxybutyric acid and acetoacetic acid excretion increases, which increases titratable acid excretion. However, because ketoacids can be metabolized to bicarbonate, their loss in the urine has no net effect on acid–base homeostasis.

ORGANIC ANION EXCRETION

Multiple organic anions in the urine can contribute to acid–base homeostasis. At least 95 different urinary organic anions have been identified, and many, including hippuric, erythronic, threonic, tartaric, and uric acids, are excreted in substantial quantities.⁷² In general, their role in acid–base homeostasis is not as a titratable acid. Instead, because their metabolism produces bicarbonate, their excretion enables alkali excretion without altering urine pH.

CITRATE EXCRETION

Citrate plays an important role in both acid–base homeostasis and preventing calcium nephrolithiasis. The latter function relates to citrate's ability to form reversible, noncovalent complexes with urinary and luminal calcium, thereby decreasing ionized calcium and decreasing the rate of calcium deposition into renal stones. Citrate may also inhibit calcium oxalate nucleation by colloidal stabilization of early-stage calcium oxalate complexes.³⁴⁰ A complete description of citrate's role in nephrolithiasis can be found elsewhere in this text. In this chapter we discuss citrate's role in acid–base homeostasis.

Citrate has two roles in acid–base homeostasis: (1) as a urinary buffer contributing to titratable acid excretion, and (2) as a substrate in the tricarboxylic acid cycle. The two primary molecular forms of citrate, citrate⁻³ and citrate⁻², exist in equilibrium with each other:



The pK_a of this buffer reaction is ~6.4. Other molecular forms, citrate⁻¹ and citrate⁰, because of the pK_a of the appropriate buffer reactions, are at such sufficiently low concentrations that they appear to not be transported to a significant extent. Thus, at a normal physiologic pH of 7.4, ~91% of total citrate is in the form of citrate⁻³ and only ~9% is citrate⁻². Because

glomerular filtrate has a pH essentially identical to systemic arterial pH, essentially all filtered citrate is in the form of citrate⁻³. In contrast, only 29% of urinary citrate at a typical urine pH of ~6.0 is in the form of citrate⁻³, meaning that ~71% has been protonated and converted to citrate⁻². This difference in citrate⁻² between filtrate and final urine enables citrate to serve as a titratable acid (see Fig. 9.6).

The second mechanism through which citrate contributes to acid–base homeostasis relates to its function as a metabolic substrate for the tricarboxylic acid cycle. Its complete metabolism, as occurs in the proximal tubule, results in HCO₃⁻ generation. Thus, citrate excretion, which is the difference between its filtration and its reabsorption, with subsequent metabolism that forms HCO₃⁻, is functionally equivalent to HCO₃⁻ excretion. Citrate excretion thereby enables base excretion without altering urine pH, which may be beneficial for minimization of pH-dependent calcium nucleation and calcium-containing stone growth.

Multiple factors regulate renal citrate excretion. Metabolic acidosis decreases and alkalosis increases citrate excretion.²⁵ Hypokalemia reduces citrate excretion.^{2,120} This effect is likely independent of systemic pH. The carbonic anhydrase inhibitor, acetazolamide, and a high dietary intake of either NaCl or protein decrease citrate excretion.^{157,213} Lithium chloride administered at therapeutic doses in animal models increases citrate excretion,⁴² but studies in humans have not confirmed this finding.⁴³

Renal tubular citrate transport is the primary determinant of citrate excretion. In humans, plasma citrate levels average ~0.1 mM, and changes in plasma levels are not an important regulatory mechanism. The proximal tubule reabsorbs a variable proportion, typically 65%–90%, of filtered citrate, and reabsorption parallels the filtered load. Citrate transported into proximal tubule cells, whether across apical or basolateral plasma membranes, is fully metabolized, enabling citrate to serve as a significant component of renal oxidative metabolism.¹⁵⁷ There does not appear to be significant transepithelial citrate transport, and there does not appear to be significant citrate transport in other renal sites.

Apical citrate transport is mediated primarily by the sodium–dicarboxylate cotransporter, NaDC1, an integral membrane protein highly expressed in the apical plasma membrane in the proximal tubule.^{223,294,357} This conclusion is based on the finding that NaDC1 expression parallels citrate reabsorption in metabolic acidosis,¹⁷ by the similarity of NaDC1 transport activity to the transport activity identified in brush border membrane vesicles,^{294,357} and by results of NaDC1 gene deletion studies. Specifically, NaDC1 deletion increases citrate excretion, along with excretion of several other Krebs cycle intermediates known to be transported by NaDC1.¹⁷¹

However, NaDC1 may not be the only protein involved in filtered citrate reabsorption. Preliminary studies report the presence of a residual citrate reabsorption process in NaDC1-knockout mice.³⁹² This additional citrate transport activity may be the calcium-regulated transport activity that has been identified in cultured proximal tubule cells.^{171,172} At present, the gene and gene product responsible for this citrate transport activity have not been identified.

Multiple mechanisms regulate proximal tubule citrate transport. First, the transported citrate form is citrate⁻², not citrate⁻³. Because the pK_a of the citrate buffer reaction is 6.4, luminal acidification, resulting from increased apical H⁺

secretion, directly increases luminal citrate⁻² concentration, which increases citrate reabsorption. Because many conditions stimulate luminal acidification, this provides a mechanism to increase filtered citrate reabsorption without altering the number or activity of citrate transporters. Second, metabolic acidosis increases apical citrate transport capacity,¹⁹⁰ most likely by increasing NaDC-1 expression.¹⁷ Both hypokalemia and starvation decrease citrate excretion, likely through stimulation of proximal tubule citrate transport.^{236,472} Several cellular signaling proteins regulate NaDC1. One is the calcineurin inhibitor target protein, cyclophilin,³³ which likely mediates the effects of calcineurin inhibitors to increase citrate reabsorption.³⁷⁵ Others include protein kinase C, sodium–hydrogen exchanger regulating factor 2, serum and glucocorticoid-inducible kinase, and protein kinase B.^{40,295} Finally, recent studies have implicated the proximal tubule basolateral bicarbonate transporter, NBCe1, as a critical determinant of NaDC1 expression.²⁹³

Basolateral citrate transport in the proximal tubule has different characteristics than apical transport. Uptake is pH-independent, Na⁺-dependent, and electroneutral, appears to involve 3 Na⁺ and 1 citrate⁻,^{157,191} and appears to be mediated by NaDC3.⁵⁹ Approximately 20% of proximal tubule citrate uptake appears to be mediated by basolateral uptake. However, because the proximal tubule does not secrete citrate, basolateral citrate uptake does not regulate renal citrate excretion.

OTHER ORGANIC ANIONS

Humans excrete 26–52 mEq d⁻¹ of organic anions other than citrate. Because organic anions can be metabolized to bicarbonate, organic anion excretion is functionally equivalent to alkali excretion and thereby can contribute to acid–base regulation. The extent of change in these organic anions with acid–base disturbances is not clear. Some studies show acid or alkali loading does not alter urinary organic anion excretion,²³⁵ whereas other studies show alkali loading increases and acid loading decreases organic anion excretion.¹⁷⁷

Quantitatively, there are important species-dependent differences in the magnitude of organic anion excretion. In humans, basal organic anion excretion averages 0.3–0.7 mEq kg⁻¹ d⁻¹,²³⁵ whereas in the rat organic anion excretion is 2–8 mEq kg⁻¹ d⁻¹.^{56,333} Studies in the dog report 1–2 mEq kg⁻¹ d⁻¹²⁹⁰ and in the rabbit average 4 mEq kg⁻¹ d⁻¹.³³³ This species-dependent variation may in part reflect differences in intestinal organic anion absorption³³³ or in the intestinal biome.

AMMONIA METABOLISM

Renal ammonia metabolism and transport is a predominant mechanism of the renal response to most acid–base disorders (see Fig. 9.5). Ammonia metabolism involves integrated function of multiple portions of the kidney. Only a minimal amount of urinary ammonia derives from glomerular filtration, making urinary ammonia excretion unique among the major compounds present in the urine. Instead, the kidney produces ammonia, which is then selectively transported either into the urine or the renal vein, where it is transported to the systemic circulation. Importantly, renal vein ammonia content exceeds arterial content, indicating the kidney is a net producer of ammonia, even when there is significant

urinary ammonia excretion. Selective ammonia transport involves integrated transport in the proximal tubule, TAL of the loop of Henle, and the collecting duct (Fig. 9.7).

AMMONIA CHEMISTRY

Ammonia exists in two molecular forms, NH₃ and NH₄⁺. The relative amounts of each are governed by the buffer reaction: NH₃ + H⁺ ↔ NH₄⁺. This reaction occurs essentially instantaneously and has a pK_a under biologically relevant conditions of ~9.15. Accordingly, the majority of ammonia is present as NH₄⁺; at pH 7.4 only ~1.7% is present as NH₃. Because most biological fluids exist at a pH substantially below this pK_a, small changes in pH cause exponential changes in NH₃ concentration, but almost no change in NH₄⁺ concentration (Fig. 9.8).

NH₃, although uncharged, has an asymmetric arrangement of positively charged hydrogen nuclei around a central nitrogen; this results in significant polarity (Fig. 9.9). As a consequence, NH₃ has limited lipid permeability. Consequently, diffusion across plasma membranes is limited, and NH₃ transporters both accelerate NH₃ transport and provide important regulatory control.

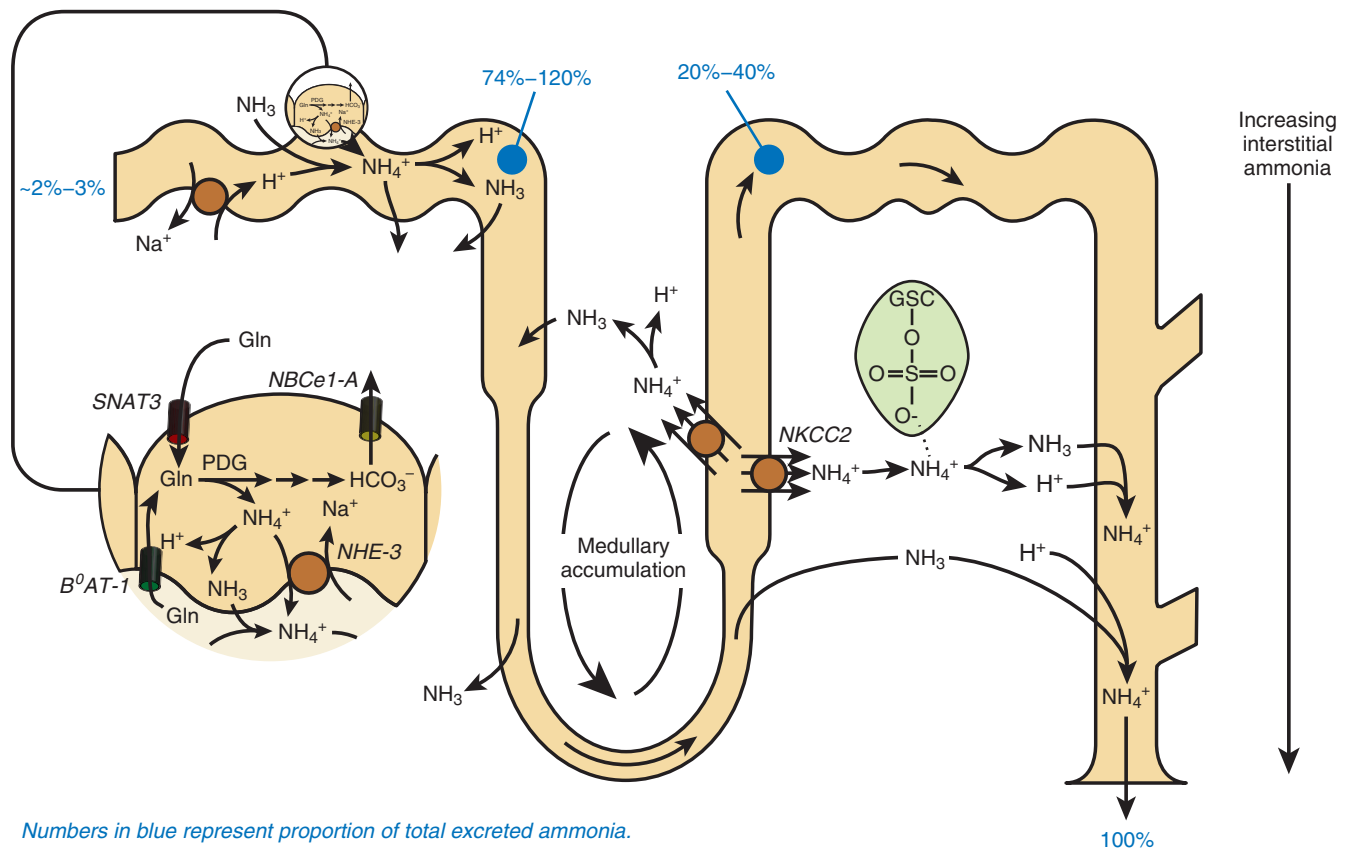
NH₄⁺ also has limited permeability across lipid bilayers in the absence of specific transport proteins. However, in aqueous solutions NH₄⁺ and K⁺ have nearly identical biophysical characteristics, which enables NH₄⁺ to be transported at the K⁺-transport site of essentially all K⁺ transporters.⁴⁵³ Several Na⁺/H⁺ exchanger (NHE) family members also appear to transport NH₄⁺ at the H⁺ binding site, resulting in Na⁺/NH₄⁺ exchange activity.

AMMONIA PRODUCTION

Almost all renal epithelial cells can produce ammonia, but the proximal tubule is the primary site for physiologically relevant ammoniogenesis.¹⁴³ The renal isoform of glutaminase (KGA), also known as phosphate-dependent glutaminase (PDG), is involved in this process.⁹¹ The proximal tubule accounts for 60%–70% of total renal ammonia production under basal conditions and at least 70%–80% in response to metabolic acidosis¹⁴³ (Fig. 9.10).

Although multiple pathways for ammoniogenesis are present in the proximal tubule (Fig. 9.11), the predominant pathway involves PDG.^{92,461} PDG is an inner mitochondrial membrane-bound enzyme that metabolizes glutamine to glutamate, producing NH₄⁺. Glutamate then undergoes further metabolism through multiple pathways. The major pathways involve glutamate dehydrogenase (GDH) with production of α-ketoglutarate (α-KG, also known as 2-oxoglutarate) and release of NH₄⁺. GDH-mediated metabolism is regulated in parallel with changes in total renal ammoniogenesis. Because glutamate is a negative regulator of PDG activity, changes in GDH activity, by changing mitochondrial glutamate levels, indirectly regulates PDG functional activity.

Glutamate can be converted back to glutamine via the enzyme glutamine synthetase. This reaction uses NH₄⁺ as a cosubstrate, decreasing net NH₄⁺ formation. Glutamine synthetase is expressed in the proximal tubule and in intercalated cells, and its expression decreases in response to metabolic acidosis^{88,226} and, in the proximal tubule, with hypokalemia.⁴²² Dietary protein restriction, which decreases ammonia excretion, increases glutamine synthetase expression, likely resulting in increased ammonia recycling and thereby



Numbers in blue represent proportion of total excreted ammonia.

Fig. 9.7 Integrated overview of renal ammonia metabolism. Renal ammoniagenesis occurs primarily in the proximal tubule, involving glutamine uptake by SNAT3 (SN1) and B⁰AT-1, glutamine metabolism forming ammonium and bicarbonate, and apical NH₄⁺ secretion involving NHE3 and parallel H⁺ and NH₃ transport. Ammonia reabsorption in the thick ascending limb, involving apical NKCC2-mediated uptake, results in medullary ammonia accumulation. Medullary sulfatides (*highlighted in green*) reversibly bind NH₄⁺, contributing to medullary accumulation. Ammonia is secreted in the collecting duct via parallel H⁺ and NH₃ secretion. Numbers in blue represent the proportion of total excreted ammonia at each location. GSC, Galactosylceramide backbone.

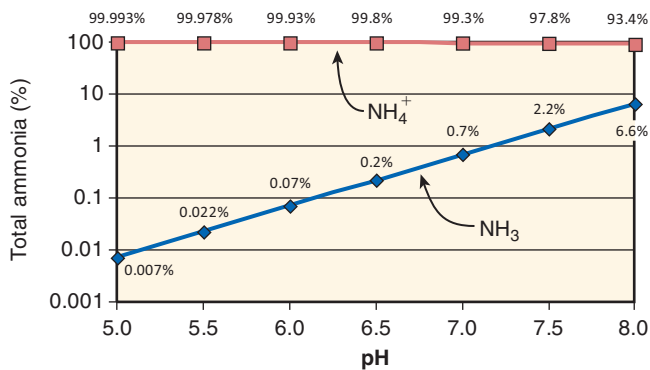


Fig. 9.8 Relative changes in NH₃ and NH₄⁺ concentration as pH changes. NH₃ and NH₄⁺ contributions to total ammonia were determined from the buffer reaction, NH₃ + H⁺ ↔ NH₄⁺. A pK_a of 9.15 was used for calculations. Amounts shown are proportion of total ammonia present as NH₃ and NH₄⁺. Note that the y-axis log transformed. Reprinted from Weiner ID, JW Verlander. Renal ammonia metabolism and transport. *Compr. Physiol.* 2013;3:201–220 with permission.

diminishing net ammoniagenesis. Proximal tubule-specific glutamine synthetase deletion blunts the decrease in ammonia excretion in response to dietary protein restriction.²²⁷ Thus, glutamine synthetase-mediated ammonia recycling, which exhibits regulation counter to that of PDG in conditions

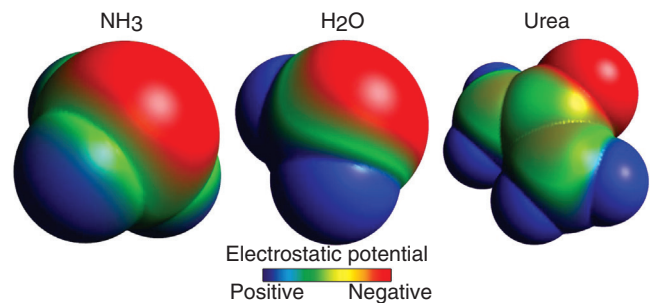


Fig. 9.9 Electrostatic charge distribution in NH₃, H₂O, and urea molecules. Models of NH₃, H₂O, and urea showing space-filling representation and surface pseudocolored to show surface charge. Each molecule, although an uncharged molecule, is polar. This polarity results in limited permeability across plasma membranes. Models generated using Avogadro software (Avogadro Chemistry, Inc.) v1.0.3.

that alter ammonia excretion, is an important component of renal ammonia metabolism.

α-KG can be metabolized through α-KG dehydrogenase and succinate dehydrogenase to form oxaloacetic acid (OAA). OAA can serve as a substrate for phosphoenolpyruvate carboxykinase (PEPCK) to form phosphoenolpyruvate (PEP), which can be used as a substrate for gluconeogenesis.

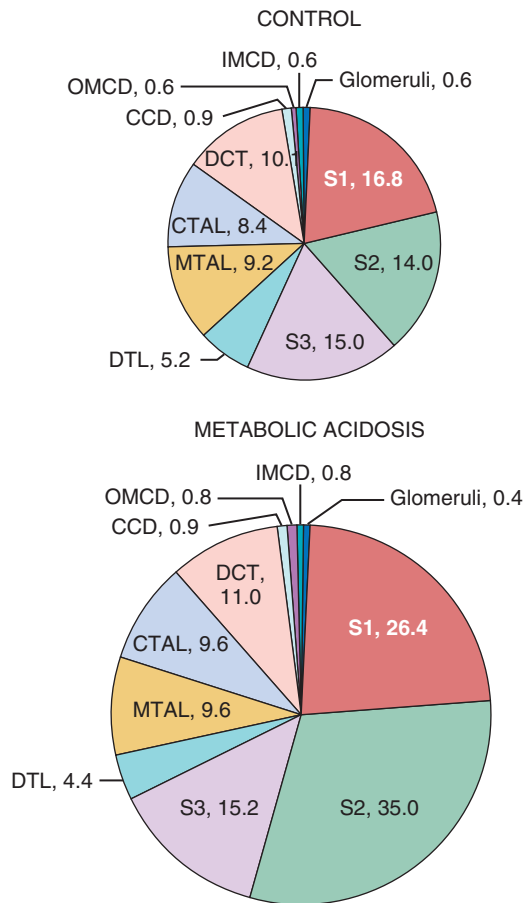


Fig. 9.10 Ammonia production in various renal segments. Ammonia production rates in different renal components measured in microdissected components from rats on control diets and after inducing metabolic acidosis. All segments tested produced ammonia. Metabolic acidosis increases total renal ammoniogenesis, but only through increased production in proximal tubule segments (S1, S2, and S3). Rates (pmol/mm) were calculated from measured ammonia production rates and mean length per segment as described in Good and Burg.¹⁴³ The size of the pie graph is proportional to total renal ammoniogenesis rates. CCD, Cortical collecting duct; CTAL, cortical thick ascending limb of Henle loop; DCT, distal convoluted tubule; DTL, descending thin limb of Henle loop; IMCD, inner medullary collecting duct; MTAL, medullary thick ascending limb of Henle loop; OMCD, outer medullary collecting duct.

Conditions that increase ammonia increase flux through this pathway and stimulate renal gluconeogenesis. Alternatively, PEP can be converted to pyruvate in an ATP-dependent reaction by pyruvate kinase. Pyruvate can then enter the TCA cycle, where its metabolism leads to ATP generation. The net result is that complete glutamine metabolism results in production of 2 NH_4^+ and 2 HCO_3^- molecules per glutamine molecule in association with a variable extent of glucose production. Conversion of PEP to pyruvate with subsequent entry into the TCA cycle results in more ATP production than utilization of PEP for gluconeogenesis.⁴⁵⁸

GLUTAMINE TRANSPORT IN AMMONIAGENESIS

Glutamine is the primary substrate for renal ammoniogenesis. Under normal acid–base balance conditions, the kidneys

extract less than 3% of the glutamine present in arterial blood flow to the kidneys. Acute metabolic acidosis induces a rapid, ~twofold increase in plasma glutamine levels; this results primarily from increased skeletal muscle and hepatic glutamine release.³⁹⁰ In parallel, renal glutamine uptake increases to as much as 20% of delivered glutamine.^{182,390} With chronic metabolic acidosis, renal extraction can increase to as much as 50% of delivered glutamine.¹⁸² Because glutamine uptake can exceed filtered glutamine, the presence of and ability to increase basolateral glutamine uptake is an important component of the regulation of ammoniogenesis.

Filtered glutamine is almost completely reabsorbed in the PCT.³⁶⁶ Multiple glutamine transporters are expressed in the apical membrane in the proximal tubule, including the Na^+ -dependent neutral amino acid transporters B⁰AT1 (SLC6A19) and B⁰AT3 (SLC6A18). Under basal conditions, luminal glutamine reabsorbed in the proximal tubule not used for ammoniogenesis can be transported across the basolateral plasma membrane. This appears to involve LAT2-4F2hc (SLC7A8-SLC3A2) and Y⁺LAT1-4F2hc (SLC7A7-SLC3A2).^{29,308,337} LAT2-4F2hc and Y⁺LAT1-4F2hc are obligatory amino acid exchangers, and the amino acid transported into the cell likely exits via basolateral TAT1 (Slc16a10), a facilitated aromatic amino acid transporter.^{29,325,326}

Basolateral glutamine uptake into proximal tubule cells appears to occur through the Na^+ -coupled, neutral amino acid transporter, SN1 (SLC38A3, also known as SNAT3).¹⁹⁶ Under basal conditions, basolateral SN1 is detectable only in the S3 proximal tubule segments, conditions that increase ammoniogenesis, such as metabolic acidosis and hypokalemia, increase S3 segment expression and induce expression in the S2 proximal tubule segment.^{62,269}

Because the initial enzyme involved in ammonia, PDG, is a mitochondrial enzyme, glutamine movement across the mitochondrial membrane is necessary. This process involves a specific transporter-mediated mechanism, is trans-stimulated and cis-inhibited by alanine, and is stimulated by metabolic acidosis.³⁴⁸ The gene and the gene product that mediate this activity are unknown at present.

AMMONIA TRANSPORT

Ammonia produced in the proximal tubule is secreted preferentially into the tubule lumen. Preferential apical secretion is due to multiple factors, including NHE3-mediated $\text{Na}^+/\text{NH}_4^+$ exchange and luminal acidification, which facilitates “trapping” of secreted NH_3 as NH_4^+ .^{275,368} However, a recent study examining the effect of proximal tubule NHE3 deletion on acid–base homeostasis found no alteration in renal ammonia excretion.²³⁸

The proximal tubule also can reabsorb luminal ammonia; this appears to occur primarily in the late proximal tubule.¹⁶⁰ These portions of the proximal tubule express glutamine synthetase, which catalyzes the reaction of NH_4^+ with glutamate to form glutamine.⁵⁷ Metabolic acidosis converts late proximal tubule ammonia transport from net reabsorption to net secretion¹⁶⁰; the molecular mechanisms that underlie this conversion involve decreased glutamine synthetase-mediated NH_4^+ metabolism.^{88,329}

The TAL reabsorbs luminal ammonia. The apical Na^+/K^+ -2 Cl^- cotransporter, NKCC2, mediates the majority of ammonia reabsorption.¹⁹ Metabolic acidosis increases both TAL ammonia reabsorption and NKCC2 expression.²⁰ Intracellular

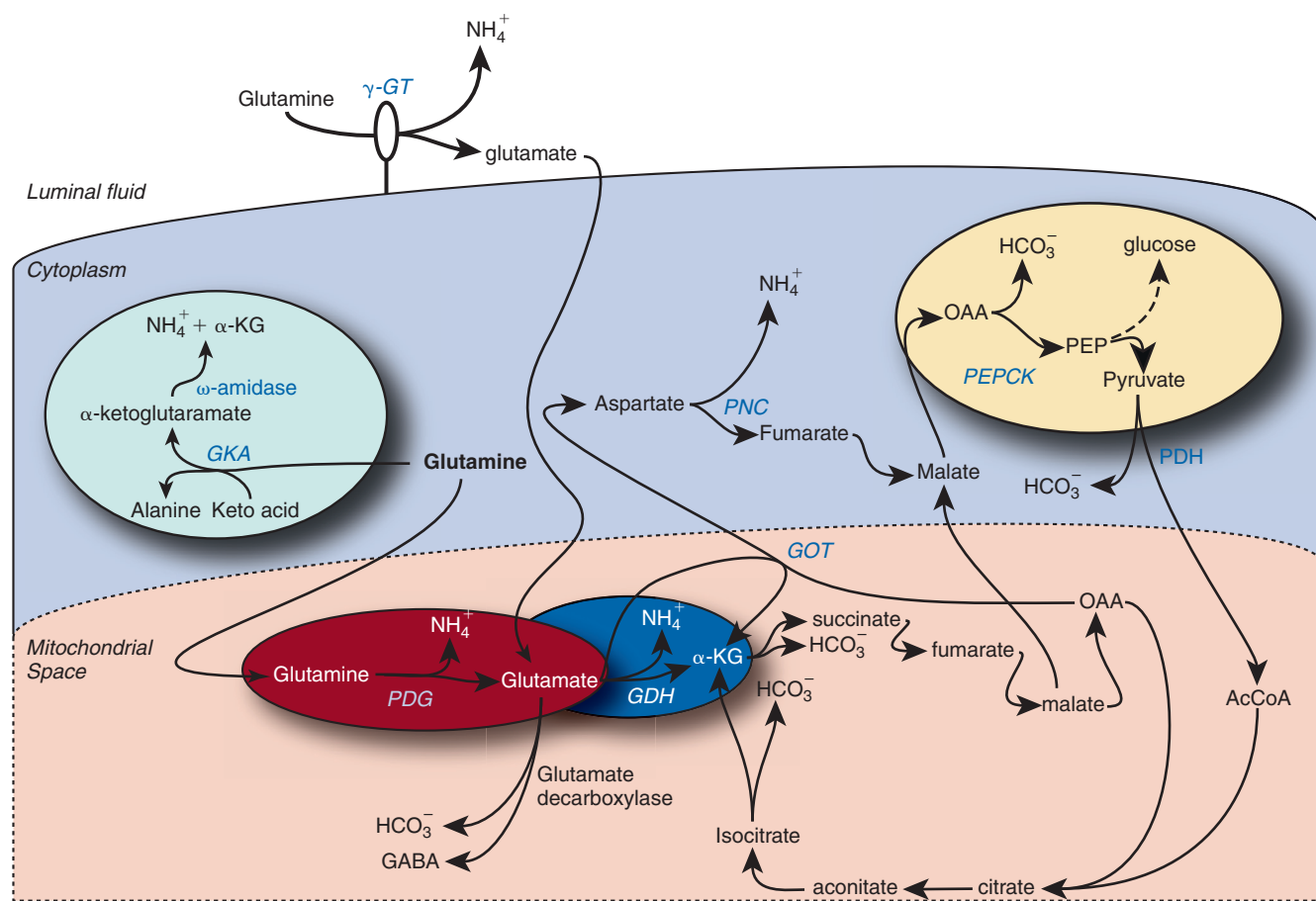


Fig. 9.11 Mechanisms of ammoniagenesis. Multiple pathways for enzymatic ammonia production originating from glutamine metabolism are present in the proximal tubule. Glutamine metabolism through PDG and GDH and involving PEPCK is the quantitatively most significant component of renal ammoniagenesis and is the primary pathway stimulated in response to metabolic acidosis. *GDH*, Glutamate dehydrogenase; *PDG*, phosphate-dependent glutaminase; *PEPCK*, phosphoenolpyruvate carboxykinase. Reprinted from Weiner ID, JW Verlander. Renal ammonia metabolism and transport. *Compr. Physiol.* 2013;201–2013 with permission.

NH_4^+ can dissociate into NH_3 and H^+ , resulting in intracellular acidification. One major mechanism of ammonia exit across the basolateral plasma membrane involves basolateral NH_4^+ transport involves NHE4-mediated $\text{Na}^+/\text{NH}_4^+$ exchange activity.^{50,461} In addition, NH_4^+ transported into the cell by NKCC2 can dissociate into NH_3 and H^+ . NH_3 appears to exit across the basolateral plasma membrane via an unidentified mechanism. Basolateral HCO_3^- entry via the electroneutral sodium–bicarbonate cotransporter, isoform 1 (NBCn1, SLC4A7), enables buffering of the released H^+ , preventing development of progressive intracellular acidosis, and facilitating continued ammonia transport.^{189,233}

Some of the ammonia absorbed by the medullary TAL undergoes recycling into the thin descending limb of the loop of Henle. This results in counter current amplification of medullary interstitial ammonia concentration. Ammonia recycling predominantly involves NH_3 transport, with a smaller component of NH_4^+ transport.¹¹⁸

The net effect of ammonia absorption by the TAL of the loop of Henle and passive ammonia secretion into the thin descending limb of the loop of Henle is development of an axial ammonia concentration gradient that parallels the hypertonicity gradient. Moreover, ammonia absorption

by the medullary TAL results in ammonia delivery to the distal tubule accounting for only ~20%–40% of final urinary ammonia content.^{104,159} Thus, tubule segments farther downstream secrete the majority of ammonia excreted by the kidneys.

There is likely to be a small component of ammonia secretion in the regions of the distal tubule prior to the CCD (i.e., the DCT, CNT, and ICT). Studies in the rat show ammonia secretion in the micropuncturable distal tubule could account for ~10%–15% of ammonia excretion.^{367,471}

Collecting duct ammonia secretion involves the integrated function of several proteins (Fig. 9.12). Several studies of the CCD, OMCD, and the IMCD have shown that collecting duct ammonia secretion involves parallel NH_3 and H^+ transport, with little-to-no transepithelial NH_4^+ permeability.^{104,206} H^+ secretion likely involves both H^+ -ATPase and H^+ - K^+ -ATPase. Carbonic anhydrase is necessary for ammonia secretion, probably through a role in supplying cytosolic H^+ for secretion.⁴³⁰ Transepithelial ammonia secretion involves both basolateral uptake and apical secretion, and specific proteins are involved in each process.

The primary basolateral ammonia transporters involved in ammonia secretion appear to differ depending on the

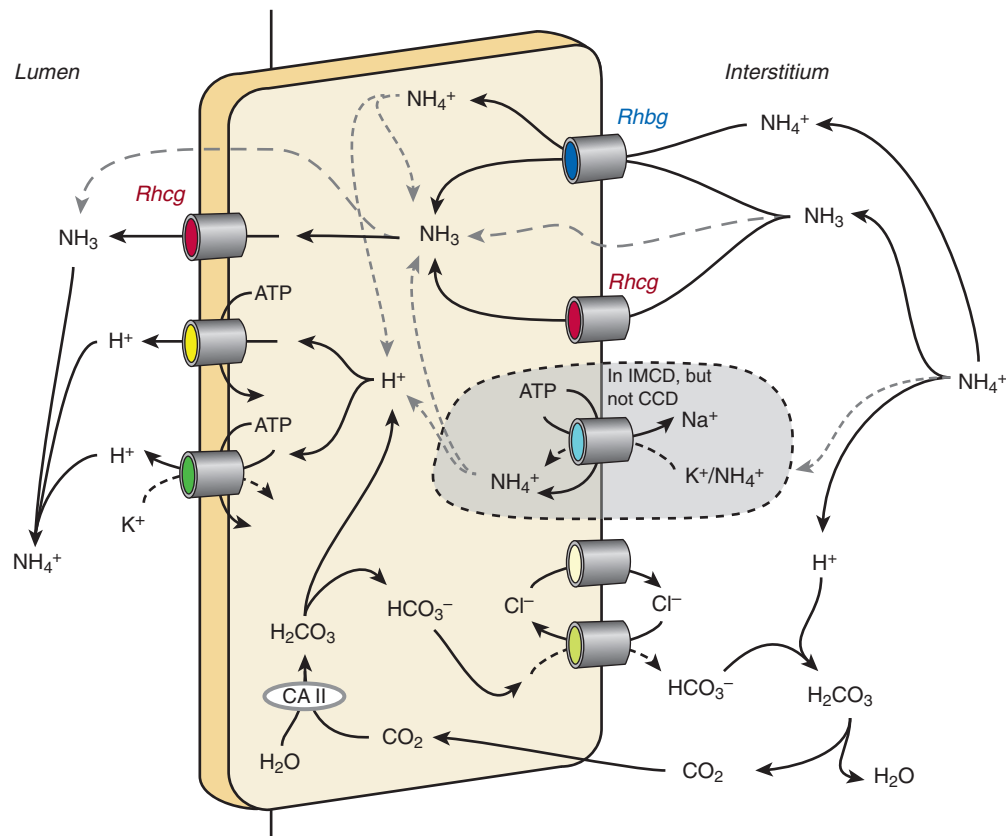


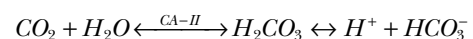
Fig. 9.12 Mechanisms of collecting duct ammonia secretion. Rhbg and Rhcg likely both contribute to basolateral ammonia uptake. As described in the text, whether Rhbg transports ammonia in the molecular form of NH_4^+ or NH_3 , or even both, remains controversial, but the likely electrochemical gradients result in uptake of both ammonia molecular species across the basolateral membrane. Rhcg mediates electroneutral NH_3 uptake across the basolateral plasma membrane. NH_3 is then secreted across the apical plasma membrane down its electrochemical gradient through processes involving both apical Rhcg transport and through a separate, undefined mechanism that may reflect lipid-phase diffusion. H^+ ions are secreted across the apical plasma membrane by both H^+ -ATPase and H^+ - K^+ -ATPase. Cytosolic H^+ is supplied either by dissociation of NH_4^+ or via a carbonic anhydrase II (CA II)-dependent bicarbonate shuttle mechanism involving basolateral chloride-bicarbonate exchange and basolateral chloride channel-mediated bicarbonate shuttling. Also shown is NH_4^+ uptake by basolateral Na^+ - K^+ -ATPase, which contributes to ammonia secretion in the IMCD where the majority of cells do not express Rh glycoproteins. IMCD, Inner medullary collecting duct.

collecting duct location. In the CCD and OMCD, the Rhesus glycoproteins, Rhbg (SLC42A2) and Rhcg (SLC42A3), appear to be the primary transport mechanisms. A hyperpolarization-activated cyclic nucleotide-gated HCN2 channel may also enable basolateral NH_4^+ uptake.⁷¹ In the IMCD basolateral Na^+ - K^+ -ATPase contributes to basolateral NH_4^+ uptake.^{428,429,438} In contrast, in the CCD inhibiting Na^+ - K^+ -ATPase does not alter ammonia secretion, suggesting either that it is uninvolved in ammonia secretion in this region of the collecting duct or that other transport mechanisms compensate when it is inhibited.²⁰⁷ Another possible NH_4^+ transport mechanism involves NKCC1. Basolateral NKCC1, which can transport NH_4^+ at the K^+ binding site, is present in both OMCD intercalated cells and IMCD cells.^{136,193} However, inhibiting NKCC1 does not alter OMCD ammonia secretion, suggesting either NKCC1 does not contribute to transepithelial ammonia secretion or that alternative transport mechanisms compensate in its absence.⁴³⁰ There may also be a component of NH_3 -gradient driven, diffusive NH_3 uptake, at least as has been identified in cultured IMCD cells.¹⁶⁵

Apical ammonia secretion appears to occur only by NH_3 movement. The primary protein involved is Rhcg. In studies

of CCD and OMCD from acid-loaded mice examined with in vitro microperfusion, Rhcg deletion decreased apical NH_3 transport by ~65%.³⁷ A significant component of the remaining NH_3 permeability may involve diffusive NH_3 transport.¹⁶⁶ Additionally, some studies have raised the possibility that, at least in response to dietary K^+ restriction, the colonic isoform of H^+ - K^+ -ATPase, HK α 2, secretes NH_4^+ .²⁸²

A substantial component of collecting duct ammonia secretion appears to involve cytosolic HCO_3^- production by CA II. Studies using microperfused collecting duct segments show that inhibiting CA-II essentially abolishes ammonia secretion.⁴³⁰ Presumably, CA II-mediated acceleration of the reaction



provides the H^+ needed for the apical H^+ secretion. The HCO_3^- is transported across the basolateral plasma membrane, in most cases likely by a basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchanger. This results either in the generation of “new” HCO_3^- or, if cytosolic NH_3 resulted from basolateral NH_3 uptake, buffering of the H^+ released from NH_4^+ as a result of the decrease in peritubular NH_3 concentration.

The absence or presence of carbonic anhydrase activity at the apical membrane also impacts collecting duct ammonia secretion. In the absence of luminal carbonic anhydrase, H^+ secretion increases the luminal $[H^+]$ above equilibrium because of relatively slow spontaneous dehydration of H_2CO_3 to CO_2 and H_2O . This is termed a luminal “disequilibrium pH.” This increased luminal acidification shifts the $H^+ + NH_3 \leftrightarrow NH_4^+$ reaction toward NH_4^+ , thereby decreasing luminal $[NH_3]$. The decreased luminal $[NH_3]$ increases the gradient for NH_3 secretion and increases net ammonia secretion. A luminal disequilibrium pH has been found and shown to accelerate ammonia secretion in the rat OMCD and terminal IMCD^{119,432} and in the rabbit CCD and OMCD^{207,376,377} but not in the rabbit OMCDi.³⁷⁶

SPECIFIC PROTEINS INVOLVED IN RENAL AMMONIA METABOLISM

Phosphate-Dependent Glutaminase

PDG is the initial enzymatic step in renal ammoniogenesis. It is located in mitochondria and catalyzes the reaction, L-glutamine + $H_2O \rightarrow$ L-glutamate + NH_4^+ . In the kidney, PDG activity is found primarily in the proximal tubule, although a lesser degree of activity is found in essentially all renal epithelial cells.^{91,197,477} The physiologic role of this activity outside of the proximal tubule is unclear. Some studies have found that metabolic acidosis increases PDG activity in the DTL, mTAL in the outer stripe, and DCT,¹⁹⁷ whereas others have found no change in activity in sites other than the proximal tubule.⁴⁷⁷ Quantitative analyses suggest that the proximal tubule is the primary site of ammonia production, and accounts for the majority of the increase in ammoniogenesis with metabolic acidosis.^{92,461}

Multiple PDG isoforms exist. In humans, the gene for the kidney-type isoform gives rise to at least two transcripts, human kidney-type glutaminase (KGA) and the glutaminase C splice variant (GAC).²⁵⁵ A separate gene gives rise to a liver-type glutaminase isoform (LGA). The KGA protein is expressed ubiquitously in the kidney, including the renal proximal tubule, and is the source of the majority of renal PDG.

Metabolic acidosis increases proximal tubule PDG activity; these increases derive from multiple mechanisms. There is increased protein expression, and this appears to be transcriptionally mediated.^{396,397} The increase in PDG mRNA results from mRNA stabilization, not increased transcription rates.¹⁸⁵ A second regulatory mechanism likely involves changes in intra-mitochondrial glutamate. Glutamate is a competitive inhibitor of PDG,³⁶⁰ and decreases in intra-mitochondrial glutamate concentration, which occur during metabolic acidosis as a consequence of increased glutamate dehydrogenase activity, increase PDG activity.³

Glutamate Dehydrogenase (GDH)

GDH is a mitochondrial enzyme that catalyzes the reaction, L-glutamate + $H_2O + NAD^+$ (or $NADP^+$) $\rightarrow$ α -KG⁻² + NH_4^+ + NADH (or NADPH) + H^+ . Two GDH isoforms exist and are products of two different genes; GLUD1 is widely expressed, including in the kidney, whereas GLUD2 appears to be a neural and testicular-specific isoform.^{257,361}

Metabolic acidosis stimulates renal GDH activity,⁴⁷⁶ both by altering its affinity for glutamate and by increasing protein and mRNA (GLUD1) expression.^{3,90,93} Acidosis also decreases

intramitochondrial α -ketoglutarate (α -KG) concentration, which contributes to increased GDH activity.³⁶⁰ Decreased α -KG accelerates GDH activity by relieving α -KG-mediated competitive inhibition of the enzymatic reaction and by inhibiting the reverse reaction.^{349,350} Changes in mRNA expression occur through changes in mRNA stability, not transcription rate.¹⁹²

Phosphoenolpyruvate Carboxykinase (PEPCK)

Renal PEPCK is a cytosolic enzyme that is the product of the PCK1 gene. In the kidney, as in extrarenal sites, including liver, adipose tissue, and small intestine, PEPCK is a key enzyme in gluconeogenesis through its role in conversion of oxaloacetate into PEP and CO_2 . It also mediates an important role in the renal response to metabolic acidosis,⁵⁸ coincident with increased renal gluconeogenesis. The adaptive increase in PEPCK activity and protein expression result from increased protein synthesis and mRNA expression.⁹⁰ In contrast to PDG and GDH, the increased PEPCK mRNA expression appears to result from increased gene transcription.¹⁶⁹

γ -GT

γ -GT accounts for phosphate-independent glutaminase activity identified in many early enzymatic studies of renal ammoniogenesis. However, γ -GT is expressed primarily in the proximal straight tubule (PST),⁹¹ and micropuncture studies suggest that glutamine is completely reabsorbed in the PCT, making PST ammoniogenesis via γ -GT unlikely to contribute significantly to renal ammoniogenesis.

NHE3

Multiple lines of evidence suggest that the apical Na^+/H^+ exchanger, NHE3, secretes NH_4^+ through binding of NH_4^+ at the H^+ binding site. These data include evidence that proximal tubule brush border membrane vesicles exhibit NH_4^+/Na^+ exchange activity,²⁰⁴ that combining a low luminal Na^+ concentration with Na^+/H^+ exchange inhibitor, amiloride, decreases ammonia secretion;²⁷⁴ and that the Na^+/H^+ exchange inhibitor, EIPA, blunts ammonia secretion when alternative secretory pathways are blocked.³⁶⁸ Against a role of NHE3 in ammonia secretion, however, is that proximal tubule-specific NHE3 deletion does not alter either basal or acidosis-stimulated ammonia excretion.²³⁸

Changes in NHE3 expression and activity during metabolic acidosis may be an important component of ammonia regulation and may be regulated by AngII and ET-1. Specific studies show that NHE3 expression and activity parallel changes in ammonia secretion in response to chronic metabolic acidosis, changes in extracellular potassium and exposure to AngII.^{16,110,276,280} In S2 and S3 segments, chronic metabolic acidosis increases AT1 receptor-mediated stimulation of NHE3.^{277,278,281} ET-1 expression increases during metabolic acidosis and subsequent activation of the ET-B receptor increases NHE3 expression and renal ammonia excretion.²²²

NHE3 is also present in the apical plasma membrane of the TAL. However, because NHE3 secretes NH_4^+ and the TAL reabsorbs NH_4^+ , NHE3 appears unlikely to mediate an important role in loop of Henle ammonia transport.

POTASSIUM CHANNELS

At a molecular level, K^+ and NH_4^+ have nearly identical biophysical characteristics. This enables essentially all K^+

transporters to also transport NH_4^+ , albeit at transport rates that are often 10%–20% of that observed for K^+ .⁴⁵³ The primary evidence that apical K^+ channels contribute to proximal tubule ammonia transport comes from in vitro microperfusion studies showing that barium, a nonspecific K^+ channel inhibitor, inhibits proximal tubule ammonia transport.³⁶⁸ Multiple K^+ channels are present in the apical plasma membrane of the proximal tubule, including KCNA10, TWIK-1, and KCNQ1; it is not known currently which of these mediates ammonia transport. In the TAL, K^+ channels can contribute to luminal NH_4^+ uptake when apical $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport is inhibited.¹⁹ However, NKCC2 inhibitors completely inhibit TAL ammonia transport, suggesting that apical K^+ channels are unlikely to mediate a quantitatively important role in TAL ammonia transport.¹⁴²

$\text{Na}^+\text{-K}^+\text{-2Cl}^-$ Cotransport

NKCC1 (SLC12A2), also known as $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter, isoform 1, is present in the basolateral plasma membrane of intercalated cells in the OMCD and IMCD and in IMCD cells.^{83,136,193} However pharmacologic inhibitors do not alter either OMCD ammonia secretion or IMCD basolateral ammonia uptake.^{429,430} Thus, NKCC1 appears unlikely to mediate a substantial role in ammonia secretion.

NKCC2 (SLC12A1), also known as $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter, isoform 2, is a kidney-specific isoform expressed in the apical plasma membrane of the TAL and is the major mechanism for ammonia reabsorption in the TAL of the loop of Henle.¹⁴² Luminal NH_4^+ competes for binding with K^+ to the K^+ -transport site, enabling alterations in luminal K^+ in hypokalemia and in hyperkalemia to alter net NH_4^+ transport and leading to alterations in medullary interstitial ammonia concentration in conditions of altered potassium homeostasis. Metabolic acidosis increases NKCC2 expression, which likely contributes to the increased ammonia reabsorption observed.^{20,138} Acidosis-induced increases in glucocorticoid levels appear to mediate increased NKCC2 expression and activity.²¹

$\text{Na}^+\text{-K}^+\text{-ATPase}$

$\text{Na}^+\text{-K}^+\text{-ATPase}$ is present in the basolateral plasma membrane of essentially all renal epithelial cells. NH_4^+ binds to and is transported at the K^+ -binding site, enabling Na^+ for NH_4^+ exchange.^{218,434} In the IMCD, $\text{Na}^+\text{-K}^+\text{-ATPase}$ -mediated basolateral NH_4^+ uptake is critical for IMCD ammonia and acid secretion.^{218,434} Decreases in interstitial K^+ levels during hypokalemia facilitate increased basolateral NH_4^+ uptake by $\text{Na}^+\text{-K}^+\text{-ATPase}$, and contribute to increased NH_4^+ secretion rates.⁴³¹ In the CCD, in contrast, basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ does not appear to contribute to CCD ammonia secretion.²⁰⁷

$\text{H}^+\text{-K}^+\text{-ATPase}$

$\text{H}^+\text{-K}^+\text{-ATPase}$ proteins are members of the P-type ATPase family and transport NH_4^+ . The majority of evidence suggests NH_4^+ is transported at the K^+ binding site, such that these proteins mediate H^+/NH_4^+ exchange. Thus, $\text{H}^+\text{-K}^+\text{-ATPase}$ is unlikely to contribute to collecting duct ammonia secretion. However, potassium deficiency increases expression of the colonic $\text{H}^+\text{-K}^+\text{-ATPase}$ and it has been postulated that this mediates increased NH_4^+ secretion via transport at the H^+ binding site.²⁸²

Aquaporins

H_2O and NH_3 have a similar molecular size and charge distribution. This has led to several studies examining whether aquaporin water channels transport NH_3 . These studies have demonstrated that some, but not all, aquaporin-family members can transport ammonia. Table 9.1 summarizes the results of these studies. With the exception of AQP8, discussed later, experimental evidence regarding the role of these proteins in renal ammonia metabolism is not available.

Substantial evidence, but not without a degree of controversy, indicates that AQP8 contributes to renal ammonia metabolism. It is found in intracellular sites in the proximal tubule, CCD, and OMCD in the kidney, but not the plasma membrane.¹¹¹ The majority of ammonia generation occurs inside mitochondria in the proximal tubule. Thus, ammonia must be transported from mitochondria to cytoplasm. Several studies using heterologous expression show AQP8 can transport ammonia^{132,175,243,344} and it is able to complement ammonia transport defects in yeast lacking endogenous ammonia transporters.¹⁸⁸ In cultured proximal tubule cells, AQP8 is located in the inner mitochondrial membrane and AQP8 knockdown decreases ammonia secretion.²⁶⁸ In vivo, metabolic acidosis increases AQP8 expression.²⁶⁸ All of these findings suggest AQP8 has an important role in proximal tubule mitochondrial ammonia transport. However, studies examining mice with AQP8 gene deletion found normal ammonia excretion, under basal conditions and following acute or chronic acid loading.⁴⁸⁰

CARBONIC ANHYDRASE

Carbonic anhydrase, in addition to its role in bicarbonate reabsorption, also contributes to ammonia secretion. Direct studies have shown that carbonic anhydrase inhibition, presumably through effects on CA II, blocks OMCD ammonia secretion.⁴³⁰ Fig. 9.12 shows the putative role of cytoplasmic CA II in facilitating transepithelial ammonia secretion.

CA IV, although functioning to increase bicarbonate reabsorption, likely decreases collecting duct ammonia secretion because it prevents a luminal disequilibrium pH. Apical CA IV expression has been demonstrated in the

Table 9.1 Ammonia Transport by Aquaporins

Aquaporin	Finding	Citation
AQP1	NH_3 transport	132, 285
	No transport	175
AQP2	No transport	132
AQP3	NH_3 transport	132
	Both NH_3 and NH_4^+ transport	175
AQP4	No transport	132, 273
AQP5	No transport	132, 273
AQP6	NH_3 transport	132
AQP7	NH_3 transport	132
AQP8	NH_3 transport	132, 344
	Both NH_3 and NH_4^+ transport	175
	Transport present, NH_3 vs. NH_4^+ not differentiated	243
AQP9	NH_3 transport	132
	NH_3 and NH_4^+ transport	175
AQP10	No transport	132

rabbit CCD type A intercalated cell, in the rabbit OMCD and IMCD,³⁵⁶ in the human CCD and OMCD,²⁴⁶ but not in the rat collecting duct.⁵³ This pattern is inconsistent with evidence of luminal disequilibrium pH in the rat CCD and OMCD,^{119,208} but is consistent with the evidence of luminal disequilibrium pH in the rabbit CCD and OMCD outer stripe segments.^{376,377}

Rh GLYCOPROTEINS

Rh glycoproteins are mammalian orthologs of Mep/AMT proteins, ammonia transporter family proteins present in yeast, plants, bacteria, and many other organisms. Three mammalian Rh glycoproteins are known: Rh A glycoprotein (RhAG/Rhag), Rh B glycoprotein (RhBG/Rhbg), and Rh C glycoprotein (RhCG/Rhcg).

Rhag/RhAG (SLC42A1)

Rhag is an essential component of the erythrocyte “Rhesus complex,” which consists of Rhag in association with RhD and RhCE subunits in what appears to be a 1:1:1 stoichiometric ratio.¹⁵⁴ Rhag transports ammonia in the form of NH_3 , whereas RhD and RhCE do not.^{254,470} In humans, RhAG deficiency leads to Rh_{null} disease, which is characterized by hemolytic anemia, spherocytosis, and lack of erythrocyte expression of RhAG, RhD, and RhCE.^{180,469} Rhag protein is present in erythrocytes and in erythrocyte precursor cells present in the bone marrow and in rodent spleen, but does not appear to be expressed in nonerythroid tissues. In particular, Rhag is not found in the kidney, except in residual erythrocytes.^{450,455}

Rhbg/RhBG (SLC42A2)

Rhbg in the kidney is found exclusively in distal epithelial cell populations, with low-level basolateral expression in the DCT, and higher-level basolateral expression in the CNT, CCD, OMCD, and IMCD.^{162,322,420} The CNT and the collecting duct have heterogeneous epithelial cell populations; type A and non-A, non-B intercalated cells, principal cells, and CNT cells express Rhbg, but expression is greater in intercalated cells. Type B intercalated cells do not express detectable Rhbg immunolabel. Although an initial study detected Rhbg mRNA but not Rhbg protein in the human kidney,⁵¹ subsequent studies identified Rhbg protein expression in a pattern nearly identical to that observed in the studies of the rat and mouse kidney.¹⁶²

The majority of evidence indicates that Rhbg contributes significantly to renal ammonia excretion during basal conditions and conditions that increase ammonia excretion. Intercalated cell-specific Rhbg deletion, while not altering ammonia excretion, induces adaptive changes in other enzymes involved in ammonia metabolism that appear to compensate for its absence, indicating a role for Rhbg under basal conditions.³⁶ Both metabolic acidosis and hypokalemia, common conditions associated with increased ammonia excretion, increase Rhbg protein expression, and genetic deletion of Rhbg from intercalated cells impairs changes in ammonia excretion in response to these stimuli.^{35,36} However, a study that used a different method of acid loading, which produced less stimulation of ammonia excretion, found no effect of Rhbg deletion.⁷⁶ This suggests that other mechanisms can compensate for the lack of Rhbg, such as those identified by Bishop et al,³⁶ if only modest increases in ammonia

Table 9.2 NH_3 and NH_4^+ Transport Characteristics of Rhbg and Rhcg

	Electroneutral		Electrogenic		Citation
	NH_3 Transport	CH_3NH_2 transport	NH_4^+ Transport	CH_3NH_3^+ Transport	
Rhbg	Absent		Present		287
		Present		Absent	248
	Present		Absent		493
		Present		Absent	252
		Present	Present	Present	283
	Absent	Present	Present	Present	286
Rhcg	Present				131
	Present		Present		65
	Present		Present		24
		Present		Absent	493
		Present		Absent	252
		Present		Absent	258
		Present		Absent	37
		Present		Absent	271
	Present		Absent		154
	Present		Absent		48
	Present				131
	Absent		Present		65

excretion are needed, whereas greater degrees of adaptation require Rhbg expression.

A number of studies have addressed the issue as to whether Rhbg transports NH_3 , NH_4^+ , or both, and have resulted in conflicting results. Table 9.2 summarizes these studies. The more recent studies suggest that Rhbg can transport both NH_3 and NH_4^+ . The molecular mechanism through which NH_4^+ is transported has not been identified, but is likely to be a form of $\text{NH}_3\text{-H}^+$ cotransport rather than direct transport of the molecular species NH_4^+ . Importantly, the electrochemical gradient across the basolateral plasma membrane is such that both electroneutral NH_3 transport and electrogenic NH_4^+ transport modes will result in ammonia transport from the interstitium into the cell cytoplasm.⁴⁶²

Rhcg/RhCG

Rhcg is expressed in the kidney exclusively in epithelial cells of distal segments.^{107,161,358,359,420} Rhcg expression is prominent in the CNT, ICT, CCD, OMCD, and IMCD, weakly expressed in late DCT cells, and exhibits both apical and basolateral expression.^{161,199,358,359} Rhcg expression differs among renal epithelial cell types. In general, type A intercalated cells express higher levels of Rhcg than do principal cells. Rhcg is not detectable by immunohistochemistry in type B intercalated cells. In the CNT, non-A, non-B intercalated cells express apical Rhcg but have very little or no basolateral Rhcg immunolabel. IMCD cells do not express detectable Rhcg.

Rhcg has an important role in renal ammonia excretion in a wide variety of conditions, including basal acid–base homeostasis, metabolic acidosis, hypokalemia, and several other conditions. Gene deletion studies show that the absence of Rhcg impairs basal ammonia excretion.^{37,224} Metabolic acidosis, hypokalemia, and high dietary protein intake increase Rhcg expression, and Rhcg expression is necessary for the

normal increase in ammonia excretion.* In contrast to high dietary protein intake, during dietary protein restriction Rhcg expression is not needed for the decreased ammonia excretion that occurs.²²⁵ With reduced renal mass there is increased single-nephron ammonia excretion, and this increase involves increased polarization of Rhcg to the apical and basolateral plasma membrane.²⁰⁰ Cyclosporine A can induce renal tubular acidosis, and this involves altered Rhcg expression.²⁴¹ Aldosterone and chronic lithium administration increase renal ammonia excretion, and both are associated with increased Rhcg expression.^{187,454} Finally, the critical role of Rhcg in collecting duct ammonia secretion has been shown in studies using in vitro microperfusion that show that Rhcg deletion, at least in CCD and OMCD from mice with chronic metabolic acidosis, impairs transepithelial ammonia secretion and impairs both apical and basolateral plasma membrane NH_3 transport.^{37,48}

Rhcg expression appears to be regulated through a variety of mechanisms. There are changes in total protein expression in a variety of conditions, as detailed earlier. In metabolic acidosis, this increase is not associated with changes in steady-state mRNA expression, indicating at least a component of posttranscriptional regulation.³⁵⁸ During a high protein diet there is a transient increase in Rhcg mRNA expression, indicating that there can also be a component of transcriptional regulation.⁴⁷ In addition, Rhcg is found in both the apical and basolateral plasma membrane and in subapical cytoplasmic vesicles, and changes in subcellular distribution and expression are a prominent component of the response to metabolic acidosis, hypokalemia, and reduced renal mass.^{35,163,200,359}

Rhcg may regulate collecting duct ammonia secretion in part through effects on H^+ -ATPase expression. Recent studies show that Rhcg and H^+ -ATPase are located within the same cellular protein complex and that Rhcg may modulate H^+ -ATPase activity and expression.⁴⁹ However, H^+ -ATPase does not appear to affect Rhcg function. This mechanism may help to coordinate ammonia and proton secretion beyond physicochemical driving forces.

Multiple studies have addressed the molecular form of ammonia that Rhcg transports. Table 9.2 summarizes the results of these studies. Essentially all studies have found that Rhcg transports ammonia in the form of NH_3 . This selective transport of NH_3 is critical for Rhcg to facilitate ammonia secretion across the apical plasma membrane in the collecting duct. Under normal conditions, the high luminal NH_4^+ concentrations present throughout the collecting duct in combination with intracellular electronegativity relative to the luminal fluid result in a substantial electrochemical gradient for electrogenic NH_4^+ transport from the lumen to the cytoplasm. Thus, if Rhcg were an electrogenic NH_4^+ transporter, it would facilitate collecting duct NH_4^+ reabsorption, and would be highly unlikely to contribute to collecting duct ammonia secretion.

CO_2 Transport by Rh Glycoproteins

Rhesus glycoproteins can transport molecules other than ammonia, specifically CO_2 . Quantitative studies using human erythrocytes deficient in RhAG show that the absence of RhAG decreases CO_2 transport.^{112,113} Studies using heterolo-

gous expression in *Xenopus* oocytes show all Rh glycoproteins can transport CO_2 .^{131,273} However, the physiologic role of Rhbg- or Rhcg-mediated CO_2 transport in the kidney is not clear. Intercalated cells use cytoplasmic CO_2 to generate, through a CA II-catalyzed process, the intracellular H^+ used for urinary acidification. Several studies using Rhbg and/or Rhcg deletion show that Rhbg and Rhcg expression are not necessary for urine acidification.[†] However, these studies cannot exclude the possibility of either altered intrarenal CO_2 concentrations, which enable diffusive CO_2 movement in the absence of Rhbg and Rhcg, or adaptive changes in other CO_2 transport mechanisms.

SULFATIDES

Sulfatides, highly charged anionic glycosphingolipids, appear to have an important role in renal ammonia metabolism. Sulfatides can reversibly bind NH_4^+ . They are expressed throughout the kidney, but levels are highest in the outer and inner medulla and metabolic acidosis increases medullary interstitial sulfatide content.³⁸⁰ Sulfatides appear to have an important role in maintaining the high inner medullary ammonia and increase in urinary acid elimination that develop during metabolic acidosis. Disruption of renal sulfatide synthesis, by a genetic approach along the entire renal tubule, led to lower urinary pH accompanied by lower ammonium excretion.³⁸⁰ After acid loading, mice deficient in renal sulfatide synthesis showed impaired ammonia excretion, decreased ammonia accumulation in the papilla, and chronic hyperchloremic metabolic acidosis.³⁸⁰ Thus, sulfatides, likely through their ability to reversibly bind interstitial NH_4^+ , have an important role in renal ammonia, handling, urinary acidification, and acid-base homeostasis.

ACID-BASE SENSORS

Several studies have begun to elucidate the molecular mechanisms through which the kidney recognizes altered systemic pH. Candidate molecular sensors have included acid-/alkali-sensing receptors, tyrosine kinases, and bicarbonate-stimulated adenylyl cyclase.

ACID-/ALKALI-SENSING RECEPTORS

GPR4

Several G-protein-coupled receptors are sensitive to extracellular pH, which results in pH-dependent intracellular cAMP or IP₃ production.²⁴⁹ Of these, GPR4 has been studied most extensively. It is expressed in the kidney, and GPR4 deletion results in mild metabolic acidosis, less acidic urine, and decreased ability to excrete an acid load.³⁸⁶ In acid-loaded mice, GPR4 deletion blunted the ability to increase expression of the intercalated cell basolateral protein AE1, and it blunted the acidosis-induced increase in type A and decrease in type B cells that were detectable in the CCD.³⁸⁵ Which renal cells express GPR4 has not been determined, but a preliminary report suggests that GPR4 is expressed in renal interstitial cells, not tubule epithelial cells.²⁶⁷

*References 37, 47, 224, 230, 358, and 359.

†References 36, 37, 76, 224, 229, 231.

INSULIN RECEPTOR–RELATED RECEPTOR (InsR-RR)

InsR-RR is a member of the insulin receptor family and may have a role regulating collecting duct acid–base transport. It is found in the kidney in the basolateral plasma membrane of type B cell and non-A, non-B intercalated cells,²⁸ and is activated by alkaline pH.⁹⁹ InsR-RR-deficient mice express lower levels of pendrin and have decreased ability to excrete an alkali load.⁹⁹

KINASES

Pyk2/ETB RECEPTOR PATHWAY

The nonreceptor tyrosine kinase, Pyk2, may function as a pH sensor in renal epithelial cells.²³⁹ In cultured proximal tubule cells, Pyk2 is activated by extracellular acidosis, and through activation of c-Src, leads to NHE3 activation.^{239,400} A parallel pathway for proximal tubule NHE3 activation exists, involving ERK1/2 and c-fos activation, but does not involve Pyk2.⁵⁵ These two pathways both increase expression of the endothelin gene, ET-1, which in turn activates the ETB receptor and increases apical plasma membrane NHE3 expression.³¹³ In cultured OMCD cells, Pyk2 is required for acidosis-induced activation of H⁺-ATPase through a signaling mechanism that involves the MAPK signaling pathway ERK1/2.¹¹⁷

RECEPTOR TYROSINE KINASE

The proximal tubule responds to changes in peritubular HCO₃[−] and CO₂ with altered rates of luminal HCO₃[−] reabsorption. This activation involves a receptor tyrosine kinase, possibly a member of the ErbB family,⁴⁸⁶ and may involve ErbB1/2 heterodimerization and activation of receptor tyrosine phosphatase-γ.⁵⁵ Acute acidosis increases tyrosine phosphorylation of ErbB1 and ErbB2, consistent with a role in a signaling cascade regulating HCO₃[−] reabsorption.³⁷⁰

BICARBONATE-STIMULATED ADENYLYL CYCLASE

Another mechanism regulating renal acid–base homeostasis involves the soluble adenylyl cyclase (sAC). Its production of cAMP is directly stimulated by increases in cytoplasmic HCO₃[−],⁸¹ and cAMP stimulates collecting duct H⁺ secretion.³⁰⁰ sAC is widely expressed in the kidney, including the TAL, DCT, and collecting duct.^{298,299} In collecting duct intercalated cells, sAC colocalizes with H⁺-ATPase in both the type A and type B intercalated cell and it coimmunoprecipitates with H⁺-ATPase, implicating it in regulation of H⁺ secretion.²⁹⁹ In clear cells of the epididymis, used as a model system of collecting duct intercalated cells, sAC regulates apical H⁺-ATPase expression through changes in cAMP production.²⁹⁸ Changes in cAMP levels also alter the expression of pendrin protein in type B and non-A, non-B intercalated cells in cultured CCD and CNT.³⁹³ Increased cAMP increases pendrin expression in these cells, whereas nitric oxide reduces pendrin expression through hydrolysis of intracellular cAMP.

DIURNAL VARIATION IN ACID EXCRETION

There is a diurnal variation in renal acid excretion, which involves ammonia and titratable acid excretion and urine

pH.³⁶⁹ Circadian changes in NHE3 expression have been identified, particularly in the TAL in the outer medulla,²⁹¹ and are likely mediated by Clock and BMAL1 genes.³⁴² There are also circadian changes in ENaC expression,³³² which by altering luminal electronegativity may alter voltage-sensitive H⁺ secretion and urine acidification. Diurnal variation in net acid excretion is altered in uric acid stone formers and may contribute to the pathogenesis of nephrolithiasis in this condition.⁶³

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BOARD REVIEW QUESTIONS

- Proximal tubule bicarbonate reabsorption increases in response to acidotic conditions. This effect occurs as a direct effect of
 - Changes in luminal fluid composition
 - Changes in intracellular pH
 - Changes in peritubular HCO_3^- and Pco_2 but not pH
 - Changes in proximal tubule cellular composition
 - Changes in extracellular K^+ , which then alter proximal tubule bicarbonate reabsorption

Answer: c

Rationale: Studies using “out of equilibrium” solutions, which enabled separate evaluation of the roles of pH, bicarbonate, and Pco_2 , show that peritubular pH does not have a direct effect on proximal tubule bicarbonate reabsorption, whereas peritubular bicarbonate and CO_2 do. Moreover, the effect of peritubular bicarbonate and CO_2 on bicarbonate reabsorption is dissociated from the effects on intracellular pH. The effect of peritubular bicarbonate and CO_2 appears to be mediated through the basolateral membrane protein, receptor protein tyrosine kinase gamma. Changes in the luminal fluid are not a major regulatory mechanism, and with decreases in the luminal bicarbonate, as occur in metabolic acidosis, a decrease bicarbonate reabsorption may indirectly stimulate peritubular bicarbonate backleak. Whereas there is increasing evidence for changes in the collecting duct cellular composition in acid–base conditions, this does not occur in the proximal tubule. Acid–base alterations in proximal tubule bicarbonate reabsorption are independent of changes in extracellular potassium concentration.

- Citrate excretion is regulated primarily through:
 - Changes in filtered load that result from changes in serum citrate levels
 - Changes in proximal tubule citrate reabsorption
 - Changes in loop of Henle citrate reabsorption
 - Changes in collecting duct citrate reabsorption
 - Changes in collecting duct citrate secretion.

Answer: b

Rationale: Citrate excretion is regulated almost entirely by changes in proximal tubule citrate reabsorption. Changes that occur in response to acid–base and electrolyte disorders are independent of plasma citrate concentrations and of filtered citrate load. There is no known regulation of citrate reabsorption in segments distal to the proximal tubule.

- The regulation of renal ammonia handling involves
 - Changes only in filtered ammonia load
 - Changes only in proximal tubule ammonia generation
 - Ammonia transport that involves only diffusive NH_3 movement and trapping
 - Specific integral membrane proteins that transport NH_3 or NH_4^+
 - Changes in ammonia recycling and glutamine regeneration in the loop of Henle

Answer: d

Rationale: A major advance in the understanding of ammonia metabolism has been the identification that specific integral membrane proteins transport NH_3 or NH_4^+ . Examples of these proteins include NHE3 in the proximal tubule, and NKCC2 and NHE4 in the thick ascending limb of the loop of Henle, and the ammonia transporter family members, Rhbg and Rhcg, in the collecting duct. This model has now essentially replaced the previous model of diffusion equilibrium for NH_3 and NH_4^+ trapping. In contrast to other urinary solutes, very little of urinary ammonia derives directly from arterial delivery; intrarenal ammonia generation is the primary source of urinary ammonia excretion. Changes in plasma ammonia levels, and thereby of filtered load for ammonia, are not a regulatory mechanism altering renal ammonia excretion. Although proximal tubule ammonia generation is involved in response to acid–base and electrolyte disorders, it is not the only regulatory mechanism, as changes in the expression and regulation of ammonia transporting proteins are also critical components of the response. Although there are changes in ammonia recycling and glutamine regeneration, which occurs via the enzyme glutamine synthetase, this occurs in the proximal tubule, and not in the loop of Henle.

- Transport of gas molecules, CO_2 , and NH_3 , that have critical roles in the maintenance of acid–base homeostasis across plasma membranes involves
 - Diffusive movement only
 - Interaction with specific lipid molecule components of the lipid bilayer
 - Peritubular transport only
 - Transport by integral membrane proteins with no selectivity between different gas molecules
 - Transport by integral membrane proteins that have varying specificity for different gas molecules.

Answer: e

Rationale: Members both of the aquaporin family and the Rhesus glycoproteins family transport both CO_2 and NH_3 . Different members have different selectivity for CO_2 or NH_3 , in some cases transporting one, but not the other. Although there is a component of diffusive movement of both CO_2 and NH_3 , increasing evidence indicates that transporter-mediated movement is a critical component of acid–base regulation. There is no evidence that interaction of CO_2 or NH_3 with specific bilayer phospholipids is a critical component of their transmembrane movement. There currently is no evidence of peritubular CO_2 or NH_3 transport in acid–base homeostasis.